

# CHAPTER 9

## SETUP AND IMPLEMENTATION OF EFFECTIVE PREDICTIVE MAINTENANCE AND CONDITION MONITORING PROGRAMS

Condition Monitoring is concerned with extracting information from machines to indicate their condition, and to enable them to be operated and maintained with safety and economy.

**BREAKDOWN MAINTENANCE:** Machines can be run until they fail, and are then repaired. This obviously crude method of operation can be very expensive in terms of lost output and machine destruction, and it may also be dangerous.

**REGULAR PREVENTIVE MAINTENANCE:** A better method is to stop machines at regular intervals for maintenance in order to reduce the chance of unplanned stoppages, i.e. breakdowns. The choice of the best regular maintenance interval presents a difficult problem, with two opposite extremes to be avoided:

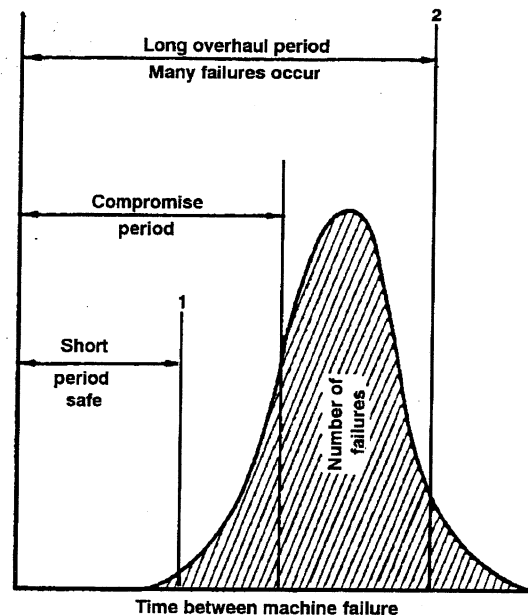
- 1) Too frequent maintenance wastes production time and increases the risk of trouble arising from human errors in reassembly:

on the other hand,

- 2) Too long an interval results in an unacceptable number of machine failures during operation.

A compromise between these two extremes can be established by experience, but machine failures will continue to occur.

**CONDITION BASED MAINTENANCE:** The best method is to carry out predictive maintenance at what may be irregular intervals but to determine these by the actual condition of the machine. Assessing the trend of vibration measurements can give a useful lead time in warning of incipient machine failure. Where considerable experience of a machine and its failure characteristics has been gained, a single check reading of its condition can provide sufficient data for determining a safe period of further running. Alternatively, an alarm indicator may be set to give warning when the safe running period has fallen to a critical level. Condition checking can also be used to compare the condition of similar machines. The readings taken may also be analyzed in detail to indicate the most likely cause of the problem.



REGULAR PREVENTIVE MAINTENANCE

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**BENEFITS OBTAINABLE BY INDUSTRY:** Benefits which can be derived from monitoring the condition of plant and machinery:

- 1) Increased plant availability resulting in greater output from the capital invested.
- 2) Reduces maintenance costs.
- 3) Improved operator and passenger safety.
- 4) More efficient plant operation and more consistent quality, obtained by matching the rate of output to the plant condition.
- 5) More effective negotiations with plant manufactures or repairers, backed up by systematic measurements of plant condition.
- 6) Better customer relations following from the avoidance of inconvenient breakdowns which would otherwise have occurred.
- 7) The opportunity to specify and design better plant in the future.

How these benefits arise:

- 1) Machine running time can be increased by maximizing the time between overhauls. Overhaul can be reduced because the nature of the problem is known, and the spares and men can be ready. Consequential damage can be reduced or eliminated.
- 2) The lead time given by condition monitoring enables machines to be stopped before they reach a critical condition, especially if instant shutdown is not permitted.
- 3) The operating load and speed on some machines can be varied to obtain a better compromise between output, and operating life to the next overhaul.
- 4) Measurements of plant when new, at the end of the guarantee period, and after overhaul, give useful comparative values.
- 5) The lead time given by condition monitoring enables such breakdowns to be avoided.
- 6) The recorded experience of the operation of the present machinery is used for this purpose.

### **ESTIMATING THE BENEFITS**

**THE BENEFITS:** Vibration monitoring techniques can be easily applied to rotating (and reciprocating) machinery to detect most common problems, such as: unbalanced, bent or misaligned shafts; bearing or gear defects; belt, blade or impeller faults; structural, foundation or critical speed problems; rubbing, loss of lubrication and oil whirl.

The main savings of using predictive maintenance for these problems arise by avoiding losses of output due to the breakdown of machinery (about 65%) and by reducing the cost of maintenance (about 35%). Output related losses can be estimated from the number of days of lost production multiplied by the gross output value per day. The maintenance costs are rather more difficult to quantify, but related mainly to the labor costs of breakdown maintenance. As a very rough initial guide, industries are likely to achieve savings on the order of 0.5 to 3% of "added value output" (gross sales minus gross costs of raw materials).

Establishments which are particularly suitable will be those in which the output is produced by machinery rather than manual work, and which have a large output from one plant, so that potential breakdown losses are high. The level of savings is also likely to be at a maximum if the plant is currently incurring high costs for maintenance (greater than 80% of averaged annual capital investment). If maintenance is already well organized and breakdown costs still exceed 40% of total maintenance costs, predictive maintenance can also prove beneficial. In industries which involve an element of risk for employees (where 0.5% of more employees are killed or seriously injured) then predictive maintenance should be regarded as essential.

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## ESTIMATING THE COSTS

**THE COSTS:** If an establishment carries out its own predictive maintenance, it will need to make an initial investment in equipment (40% of total) as well as training and experience acquisition (60% of total). There is also the ongoing costs of operating the program.

As a general guide for an industrial plant, a reasonable level of initial investment is 1% of total capital value of the plant monitored (5% for special safety considerations). The ongoing costs of operating a program depend on the approach but can be quite low if done on a small scale by existing personnel. The average cost is about 16% of gross savings. Establishments large enough to employ at least one person full time on the program will realize more success, because expertise at interpreting trends and readings can be maintained. Outside services for training personnel and initially setting up the program can prove financially wise, especially for small plants (fewer than 100 employees). This is because start-up costs are high and benefits long term. Upper management should be prepared to (1) put in the effort needed to organize and refine the predictive maintenance and (2) meet its costs for 1-2 years until the benefits become apparent.

## FINAL ANALYSIS

**SUMMARY:** An accurate analysis of the costs and benefits of predictive maintenance should convincingly indicate its technical and economic importance to plant users and equipment manufacturers. The typical cost/benefit ratio of a well run program is **about \$5.00 savings for every \$1.00 expended**. Such profitability demands the interest of anyone concerned with modern industry and machinery.

**REFERENCES AND CREDITS:** This overview is based on:

- 1) A Survey of Condition Monitoring in British Industry and its Potential for wider Application. Michael Neale and Associates Report TRD 187 for the Department of Industry. June 1975.
- 2) An Updating of Report TRD 187 (1975) to Determine the Potential in 1978 for the Application of Condition Monitoring in British Industry. Michael Neale and Associates Report TRD 223 for the Department of Industry. April 1978.

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**EXAMPLE DOCUMENTED SAVINGS FROM A SUCCESSFUL PREDICTIVE MAINTENANCE  
PROGRAM OPERATED BY TECHNICAL ASSOCIATES  
(Report Excerpt)**

**TOTAL PMP PROGRAM COST SAVINGS DURING 1988-89:**

Table I summarizes cost savings achieved during the 1988-89 contract in each of the major categories considered - maintenance labor and material savings; electrical energy cost savings; and production savings. It shows that during the first ten months, the total savings due only to maintenance costs and the electrical energy costs have been approximately \$180,171 to date. This has been achieved with an investment of \$44,580 for the full year to Technical Associates. Therefore, based on only the ten months to date, this represents a benefit/cost ratio of over 4.04; if the cost savings rate during the first ten months continues through the full year, Table I shows that these savings will total about \$216,205, or a benefit/cost ratio of almost 4.85. If production savings are taken into account, this ratio should substantially increase to over 10 to 1.

**TABLE I  
ESTIMATED PMP PROGRAM COST SAVING  
FOR 1987-88 and 1988-89 CONTRACTS**

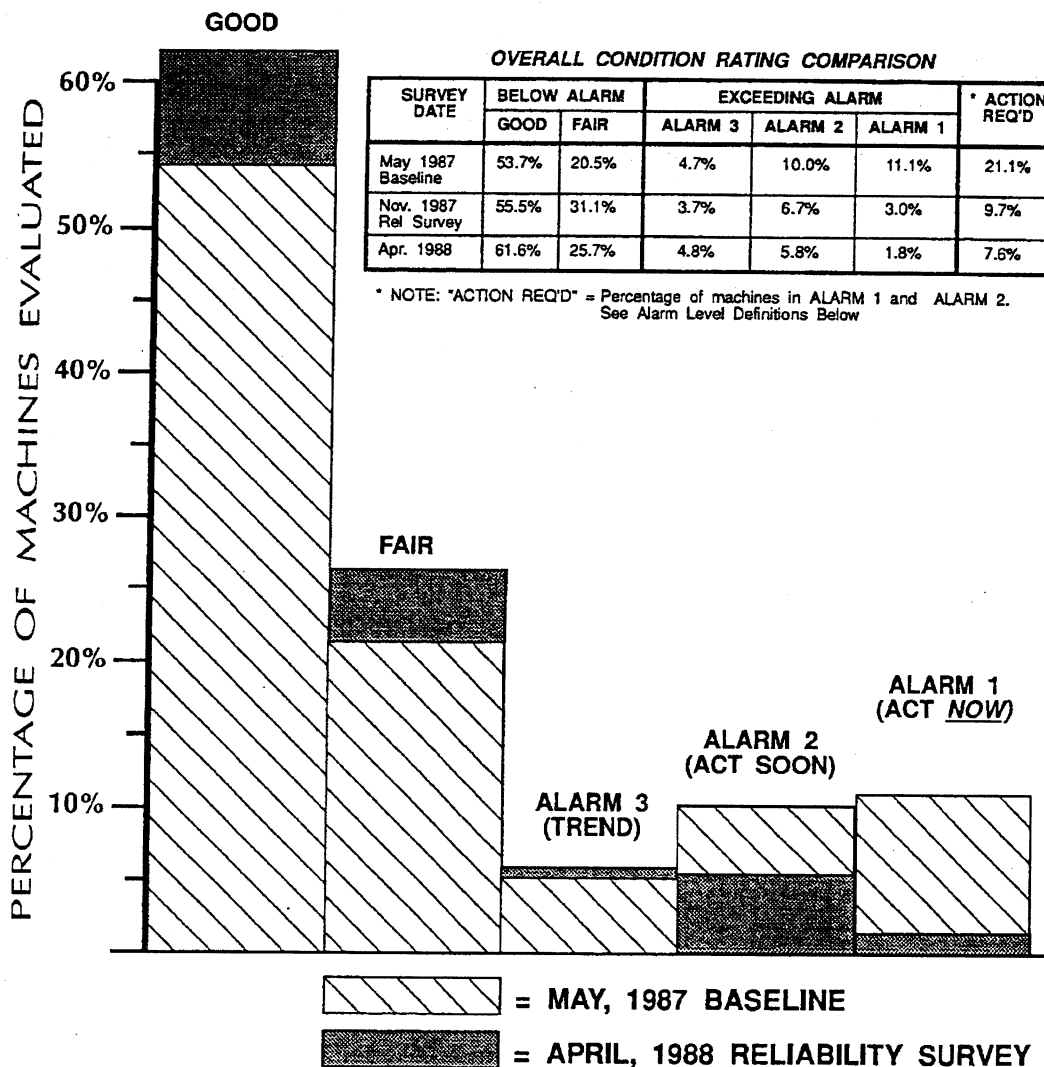
| SAVINGS CATEGORY                                 | 1987-88           | 1988-89 SAVINGS    |                    |
|--------------------------------------------------|-------------------|--------------------|--------------------|
|                                                  |                   | FIRST<br>10 MONTHS | ANNUALIZED<br>RATE |
| A. Maintenance Labor & Material Savings          | \$ 86,496         | \$ 134,385         | \$ 161,262         |
| B. Electrical Energy Cost Savings                | \$ 28,389         | \$ 45,786          | \$ 54,943          |
| SUBTOTAL                                         | \$ 114,885        | \$ 180,171         | \$ 216,205         |
| C. Production Saving<br>(Increased Availability) | \$ 172,992        | \$ 268,770         | \$ 322,524         |
| <b>TOTAL ESTIMATED CONTRACT SAVINGS</b>          | <b>\$ 287,877</b> | <b>\$ 448,941</b>  | <b>\$ 538,729</b>  |
| D. Investment with Technical Associates          | \$ 49,368         | \$ 37,150          | \$ 44,580          |

**CLOSING**

We at Technical Associates are very pleased with the results to date on our 1988-89 Predictive Maintenance contract. We feel strongly this has been due to the excellent joint cooperation between our two companies including the continuing action you have taken during the year when monthly recommendations were provided by Technical Associates. Frankly, we consider this program at your company our **"Model Predictive Maintenance Program"** and would hope that all our clients would achieve this magnitude of savings by the constant reaction of their maintenance departments to recommendations offered in the **PMP** reports.

# FIGURE 1

## COMPARISON OF OVERALL CONDITION RATING SINCE IMPLEMENTATION OF PLANT PREDICTIVE MAINTENANCE PROGRAM (For Machines in Areas 5, 6, and 7)



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**ATTACHMENT II.**  
**POTENTIAL ANNUAL ELECTRICAL ENERGY SAVINGS FROM PREDICTIVE MAINTENANCE PROGRAM**

**ASSUMPTIONS:**

1. Assuming 1.6% Reduction in Motor Current Driving Load due to decreased vibration-induced dynamic loading (from unbalance, misalignment, worn components and other problems which were detected and corrected). Current National Average shows approximately 1.6% to 2.4% energy reduction for those plants which fully implement Corrective Condition Monitoring Programs.
2. Assuming 4.6 c/kWhr Electricity Cost
3. Assuming 171 electrical motors with average capacity of 66.83 HP for those machines now in program (from actual survey of HP of each motor).
4. Assuming 24 hr/day operation year-round

**A. ANNUAL ENERGY COSTS REQUIRED FOR 1 MOTOR OF 66.83 HP RUNNING 24 HR/DAY (8760 HRS/YR) AT \$.046/kWH:**

$$= (66.83 \text{ HP}) \left( \frac{.7457 \text{ kw}}{\text{HP}} \right) \left( \frac{8760 \text{ hrs}}{\text{yr}} \right) \left( \frac{$.046}{\text{kWhr}} \right) = \underline{\$20,082} \text{ per motor}$$

**B. ANNUAL ENERGY COSTS FOR 171 MOTORS OF 66.83 HP AVERAGE SIZE:**

$$= \frac{(\$20,082)}{\text{yr-motor}} (171 \text{ motors}) = \underline{\$3,434,022} \text{ (for 171 motors)}$$

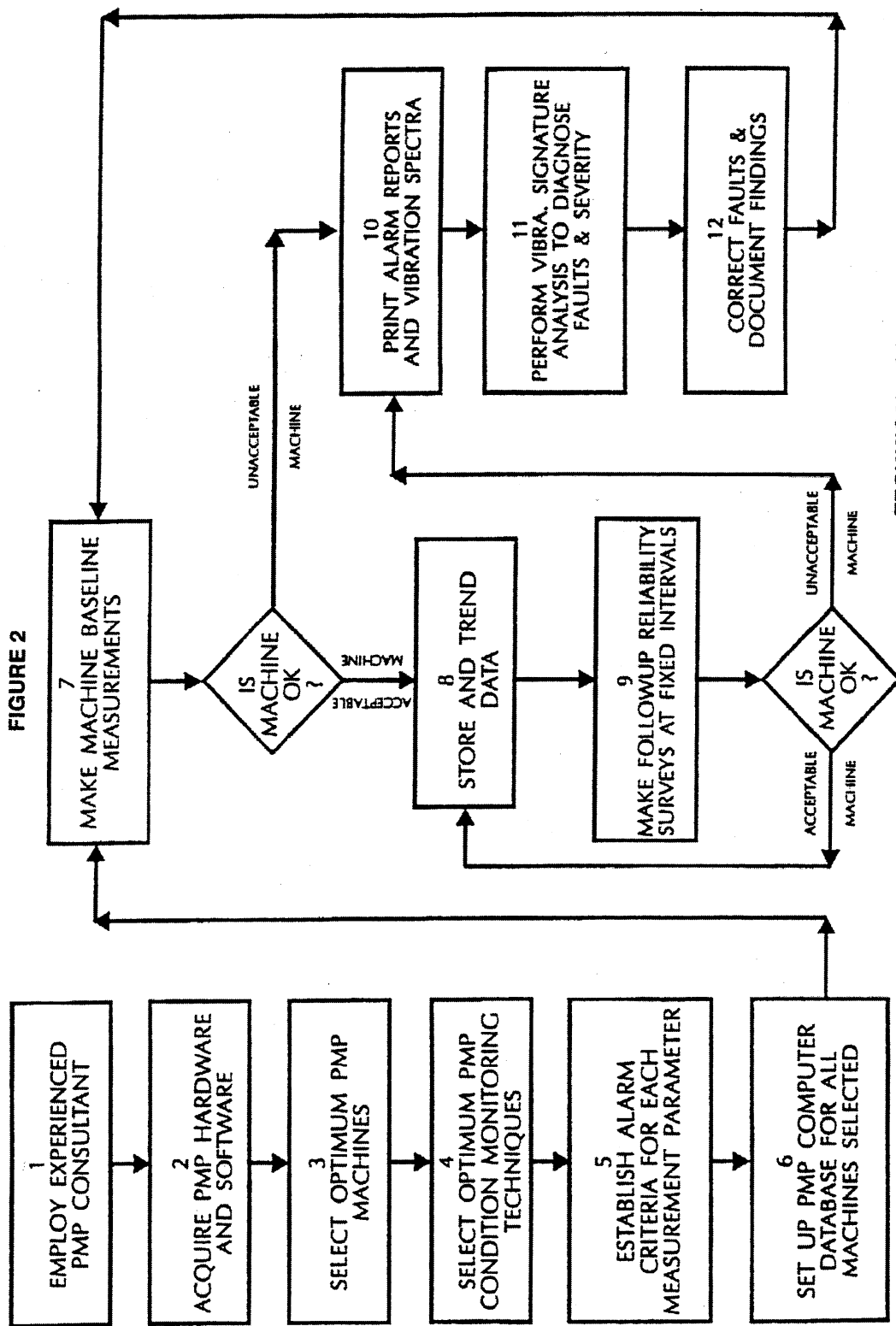
$$(74,651,033 \text{ kWhr/yr})$$

**C. PMP ENERGY SAVINGS FOR A 1.6% MOTOR CURRENT REDUCTION:**

$$= \frac{(\$3,434,022)}{\text{yr}} (.016) = \underline{\$54,943/\text{yr}} \text{ (Estimated Annual Electrical Energy Savings on 171 PMP Motors)}$$

$$(\$45,786 \text{ during first 10 months of year})$$

# RECOMMENDED PREDICTIVE MAINTENANCE PROGRAM FLOW CHART



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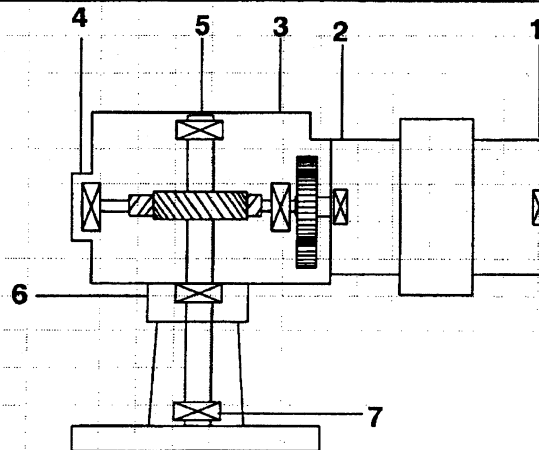




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# MACHINERY DATA GUIDE SHEET

**RELIABILITY SURVEY** predictive maintenance program (PAGE \_\_\_ OF \_\_\_)



**MACHINE NAME:** #1 AGITATOR PRO1 **BLDG.:** 124  
**COMPUTER NAME:** \_\_\_\_\_ **FLM #:** 269PA06000

| DRIVER              | INTERMEDIATE                | DRIVEN            |
|---------------------|-----------------------------|-------------------|
| RPM min: 1760       | RPM min:                    | RPM min:          |
| max:                | max:                        | max:              |
| MFGR:               | RATIO: (1.8:1 10:1) 18:1    | MFGR: PFAUDLER    |
| MODEL:              | MODEL: FMDWV50600-EJD (5RW) | MODEL:            |
| SER NO.:            | SER NO.: 88099165-002       | SER NO.:          |
| Volts: Amps: Phase: | RATING:                     | RATING:           |
| SHEAVE PITCH DIA.:  | BELT LENGTH:                | #BLADES OR VANES: |
| *BRG#: (IB)         | *BRG#: (IB)                 | *BRG#: (IB)       |
| *BRG#: (OB)         | *BRG#: (OB)                 | *BRG#: (OB)       |
| HP: 25              | #INPUT GEAR TEETH:          | COUPLING TYPE:    |
| #ROTOR BARS:        | #OUTPUT GEAR TEETH:         | OTHER:            |

**\*NOTE:** If Bearing is a Plain (or Sleeve) Bearing, simply enter "Sleeve" where shown.

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**FIGURE 4**

## TABLE II TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

CRITERIA FOR OVERALL CONDITION RATING (PEAK OVERALL VELOCITY, IN/SEC)\*

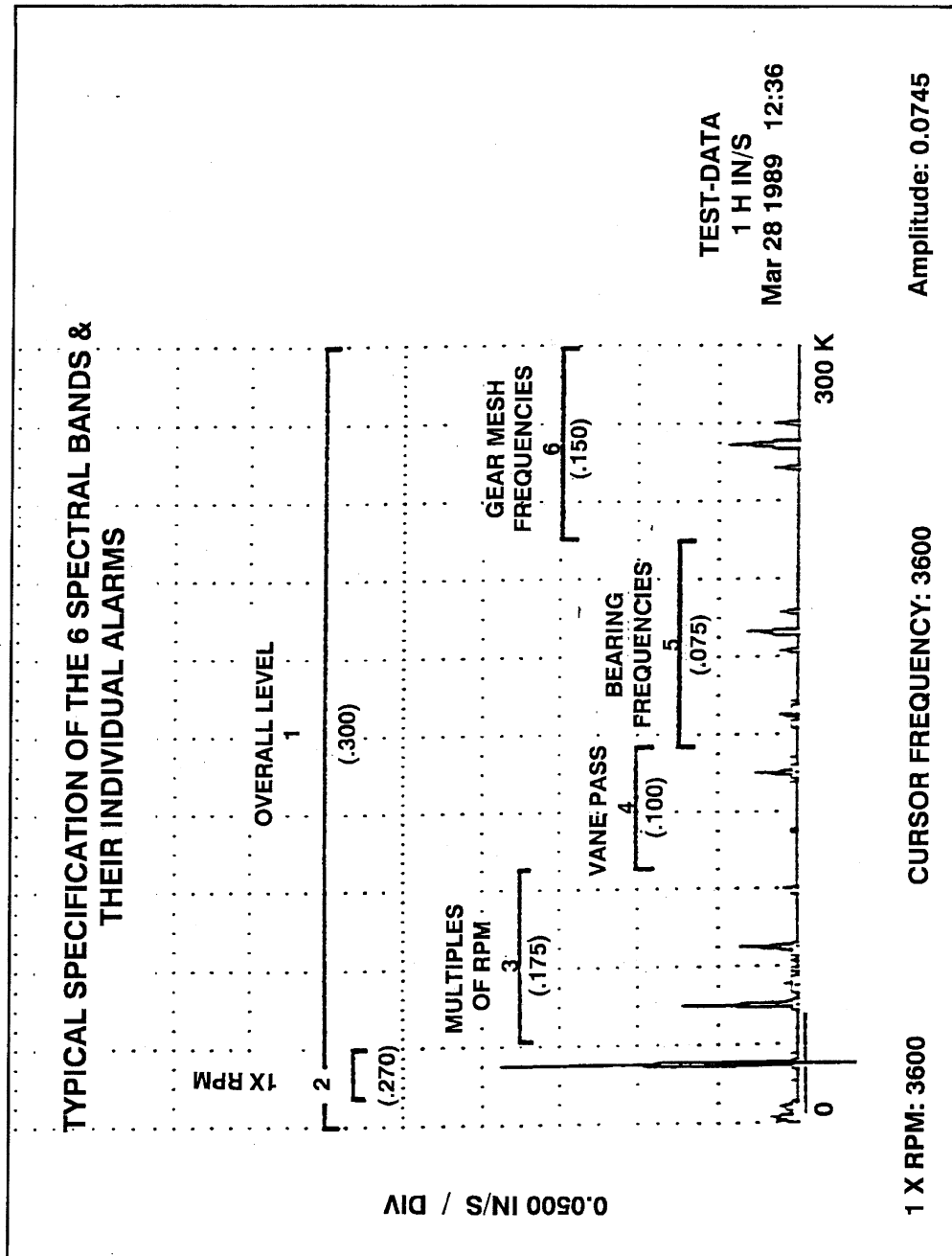
1. Assuming Machine Speed = 600 to 60,000 RPM.
2. Assuming Measurements by Accelerometer or Velocity Pickup securely mounted as Close as Possible to Bearing Housing.
3. Assuming Machine Is Not Mounted on Vibration Isolators (for Isolated Machinery - Set Alarm 30% to 50% Higher).
4. Set Motor Alarms the Same as that for the Particular Machine Type unless Otherwise Noted.
5. Consider Setting Alarms on Individual External Gearbox Position about 25% Higher than that for a particular Machine Type.

| MACHINE TYPE                                                      | GOOD     | FAIR        | (WARNING) | (FAULT) |
|-------------------------------------------------------------------|----------|-------------|-----------|---------|
|                                                                   |          |             | ALARM 1   | ALARM 2 |
| <b>COOLING TOWER DRIVES</b>                                       |          |             |           |         |
| Long, Hollow Drive Shaft                                          | 0 - .375 | .375 - .600 | .600      | .900    |
| Close Coupled Belt Drive                                          | 0 - .275 | .275 - .425 | .425      | .650    |
| Close Coupled Direct Drive                                        | 0 - .200 | .200 - .300 | .300      | .450    |
| <b>COMPRESSORS</b>                                                |          |             |           |         |
| Reciprocating                                                     | 0 - .325 | .325 - .500 | .500      | .750    |
| Rotary Screw                                                      | 0 - .300 | .300 - .450 | .450      | .650    |
| Centrifugal With or W/O External Gearbox                          | 0 - .200 | .200 - .300 | .300      | .450    |
| Centrifugal - Integral Gear (Axial Meas.)                         | 0 - .200 | .200 - .300 | .300      | .450    |
| Centrifugal - Integral Gear (Radial Meas.)                        | 0 - .150 | .150 - .250 | .250      | .375    |
| <b>BLOWERS (FANS)</b>                                             |          |             |           |         |
| Lobe-Type Rotary                                                  | 0 - .300 | .300 - .450 | .450      | .675    |
| Belt-Driven Blowers                                               | 0 - .275 | .275 - .425 | .425      | .650    |
| General Direct Drive Fans (with Coupling)                         | 0 - .250 | .250 - .375 | .375      | .550    |
| Primary Air Fans                                                  | 0 - .250 | .250 - .375 | .375      | .550    |
| Vacuum Blowers                                                    | 0 - .200 | .200 - .300 | .300      | .450    |
| Large Forced Draft Fans                                           | 0 - .200 | .200 - .300 | .300      | .450    |
| Large Induced Draft Fans                                          | 0 - .175 | .175 - .275 | .275      | .400    |
| Shaft-Mounted Integral Fan (Extended Motor Shaft)                 | 0 - .175 | .175 - .275 | .275      | .400    |
| Vane-Axial Fans                                                   | 0 - .150 | .150 - .250 | .250      | .375    |
| <b>MOTOR/GENERATOR SETS</b>                                       |          |             |           |         |
| Belt-Driven                                                       | 0 - .275 | .275 - .425 | .425      | .675    |
| Direct Coupled                                                    | 0 - .200 | .200 - .300 | .300      | .450    |
| <b>CHILLERS</b>                                                   |          |             |           |         |
| Reciprocating                                                     | 0 - .250 | .250 - .400 | .400      | .600    |
| Centrifugal (Open-Air) - Motor & Compressor Separate              | 0 - .200 | .200 - .300 | .300      | .450    |
| Centrifugal (Hermetic) - Motor & Impellers Inside                 | 0 - .150 | .150 - .225 | .225      | .350    |
| <b>LARGE TURBINE/GENERATORS</b>                                   |          |             |           |         |
| 3600 RPM Turbine/Generators                                       | 0 - .175 | .175 - .275 | .275      | .400    |
| 1800 RPM Turbine/Generators                                       | 0 - .150 | .150 - .225 | .225      | .350    |
| <b>CENTRIFUGAL PUMPS</b>                                          |          |             |           |         |
| Vertical Pumps (12' - 20' Height)                                 | 0 - .325 | .325 - .500 | .500      | .750    |
| Vertical Pumps (8' - 12' Height)                                  | 0 - .275 | .275 - .425 | .425      | .650    |
| Vertical Pumps (5' - 8' Height)                                   | 0 - .225 | .225 - .350 | .350      | .525    |
| Vertical Pumps (0' - 5' Height)                                   | 0 - .200 | .200 - .300 | .300      | .450    |
| General Purpose Horizontal Pump - Direct Coupled                  | 0 - .200 | .200 - .300 | .300      | .450    |
| Boiler Feed Pumps - Horizontal Orientation                        | 0 - .200 | .200 - .300 | .300      | .450    |
| Piston Type Hydraulic Pumps - Horizontal Orientation (under load) | 0 - .150 | .150 - .250 | .250      | .375    |
| <b>MACHINE TOOLS</b>                                              |          |             |           |         |
| Motor                                                             | 0 - .100 | .100 - .175 | .175      | .250    |
| Gearbox Input                                                     | 0 - .150 | .150 - .225 | .225      | .350    |
| Gearbox Output                                                    | 0 - .090 | .090 - .150 | .150      | .225    |
| Spindles:                                                         |          |             |           |         |
| a. Roughing Operations                                            | 0 - .065 | .065 - .100 | .100      | .150    |
| b. Machine Finishing                                              | 0 - .040 | .040 - .060 | .060      | .090    |
| c. Critical Finishing                                             | 0 - .025 | .025 - .040 | .040      | .060    |

\*NOTE: The "ALARM 1" and "ALARM 2" overall levels given above apply only to in-service machinery which has been operating for some time after initial installation and/or overhaul. They do not apply (and are not meant to serve as) Acceptance Criteria for either new or rebuilt machinery.

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**FIGURE 5**

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## FIGURE 6 DESCRIPTION OVERALL CONDITION RATING SUMMARY CHART

An overall condition rating of each machine covered under the **PREDICTIVE MAINTENANCE PROGRAM** is provided for quick reference. This condition rating is determined by rigorous review of **Reliability Survey** vibration data including trends of overall levels, trends of selected key frequencies, analysis of vibration spectra and review of "waterfall spectra" (showing how vibration signatures change from one survey to the next).

Next, an assessment is made of the severity of any problems detected during the latest **Reliability Survey**. Both the problem found and the recommended solutions are included in the "**Rank-Ordered Summary of Results and Recommendations for Problem Machines**" chart. On this chart, the severity of each problem and when it should be resolved is listed in order of highest to lowest problem severity.

Each of these levels of recommended action are then repeated in this "Overall Condition Rating Summary Chart" with headings of "Trend Problem Only"; "Schedule Repair at Convenience"; and "Maintenance Required ASAP". Nothing needs to be done by plant maintenance at this time for those machines listed under "Trend Problem Only". Simply refer to the problem detected and watch for any deterioration of this problem in future **Reliability Surveys**.

It is strongly recommended that maintenance action be taken at the earliest possible moment for those machines listed under the column "Maintenance Required ASAP". These machines have been found to have serious problems which might cause catastrophic failure in the near future. Recommended actions are listed in the aforementioned results and recommendations chart.

Maintenance should be scheduled when possible for those machines listed under "Schedule Repair at Convenience". In general, these machines have less serious problems at the time of the Reliability Survey, but still mandate close attention for any possible deterioration. Again, recommended actions are listed in the results and recommendations summary chart.

AS OF DECEMBER 8, 1992

| MACHINE<br>LOCATION (TRAIN) | MACHINES NOT IN ALARM                                                                                                                                                                                                                                  |                                    | MACHINES WITH PROBLEMS DETECTED (IN ALARM)                                                                                                                                                                                                                          |                                                   |                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------|
|                             | GOOD                                                                                                                                                                                                                                                   | FAIR                               | Trend Problem Only<br>(Priority 3)                                                                                                                                                                                                                                  | Schedule Repair at<br>Convenience<br>(Priority 2) | Maintenance<br>Required ASAP<br>(Priority 1) |
| BLD-101A                    | 101A ABS AH-21<br>101A RETN AHU-6<br>101A RETN AHU-7                                                                                                                                                                                                   | 101A ABS AH-19                     | 101A ABS AH-22<br>101A SUPY AHU-6<br>101A SUPY AHU-7<br>SLA BLOWER 1A                                                                                                                                                                                               | 101A ABS AH-20<br>1C-CR BLOWER                    | 101A ABS AH-23                               |
| BLD-101B                    |                                                                                                                                                                                                                                                        |                                    | HEATREAT VPMP-1                                                                                                                                                                                                                                                     |                                                   |                                              |
| BLD-101C                    |                                                                                                                                                                                                                                                        |                                    | 101C RETN AHU-8<br>101C SUPY AHU-8<br>101C SUPY AHU-9                                                                                                                                                                                                               | 101C RETN AHU-9                                   |                                              |
| BLD-102A                    |                                                                                                                                                                                                                                                        | 102A SUPY AH-31<br>102A SUPY AH-7  | 102A SUPY AH-25<br>102A SUPY AH-26<br>102A SUPY AH-32<br>102A SUPY AH-33<br>102A SUPY AH-35                                                                                                                                                                         | 102A SUPY AH-34<br>102A SUPY AH-36                |                                              |
| BLD-102B                    |                                                                                                                                                                                                                                                        | 102B SUPY AH-41<br>102B SUPY AH-43 | 102B SUPY AH-38<br>102B SUPY AH-40<br>102B SUPY AH-42                                                                                                                                                                                                               |                                                   | 102B SUPY AH-44                              |
| BLD-102C                    |                                                                                                                                                                                                                                                        | 102C SUPY AH-51<br>102C SUPY AH-52 | 102C SUPY AH-50<br>102C SUPY AH-53<br>102C SUPY AH-54                                                                                                                                                                                                               |                                                   |                                              |
| BLD-103                     | 103 AHUSYS P301<br>103 AHUSYS P302<br>103 COMPC L P303                                                                                                                                                                                                 | CAFETER AHU-309                    | 103 COMPC L P304<br>103 RCP CHILR 1<br>103 RCP CHILR 2<br>103B COMPRM GEN                                                                                                                                                                                           | CAFETER AHU-308                                   |                                              |
| BLD-105                     | #3 CHILR SUP P7<br>#3 YORK CHILLER<br>#6 CHIL SUP P10<br>CENTAC COMP 4AF<br>CH WTR CIRC P12<br>COMP COOLING P26<br>COMP COOLING P27<br>COOL TWR VPMP17<br>COOL TWR VPMP18<br>COOL TWR VPMP19<br>COOLING TOWER 02<br>VS CWTR CRC P11<br>VS HOTWATER P-1 | CENTAC COMP-2<br>CENTAC COMP-3     | #6 YORK CHILLER<br>CENTAC COMP-1AF<br>CH WTR CIRC P13<br>COND WTR VPMP22<br>COND WTR VPMP23<br>COND WTR VPMP24<br>COOL TWR VPMP20<br>COOL TWR VPMP21<br>COOLING TOWER 3A<br>COOLING TOWER 3C<br>DIESEL ENG PUMP<br>DIESEL P-1<br>EMERG DIESEL GEN<br>HOT WATER P-28 | COOLING TOWER 01<br>COOLING TOWER 3B              | COND WTR VPMP25                              |
| INCIN                       | INCIN VA FAN                                                                                                                                                                                                                                           |                                    | INCIN HYD PMP 1                                                                                                                                                                                                                                                     |                                                   |                                              |

For Various Reasons, The Following Machines Were Not Operating or Measured

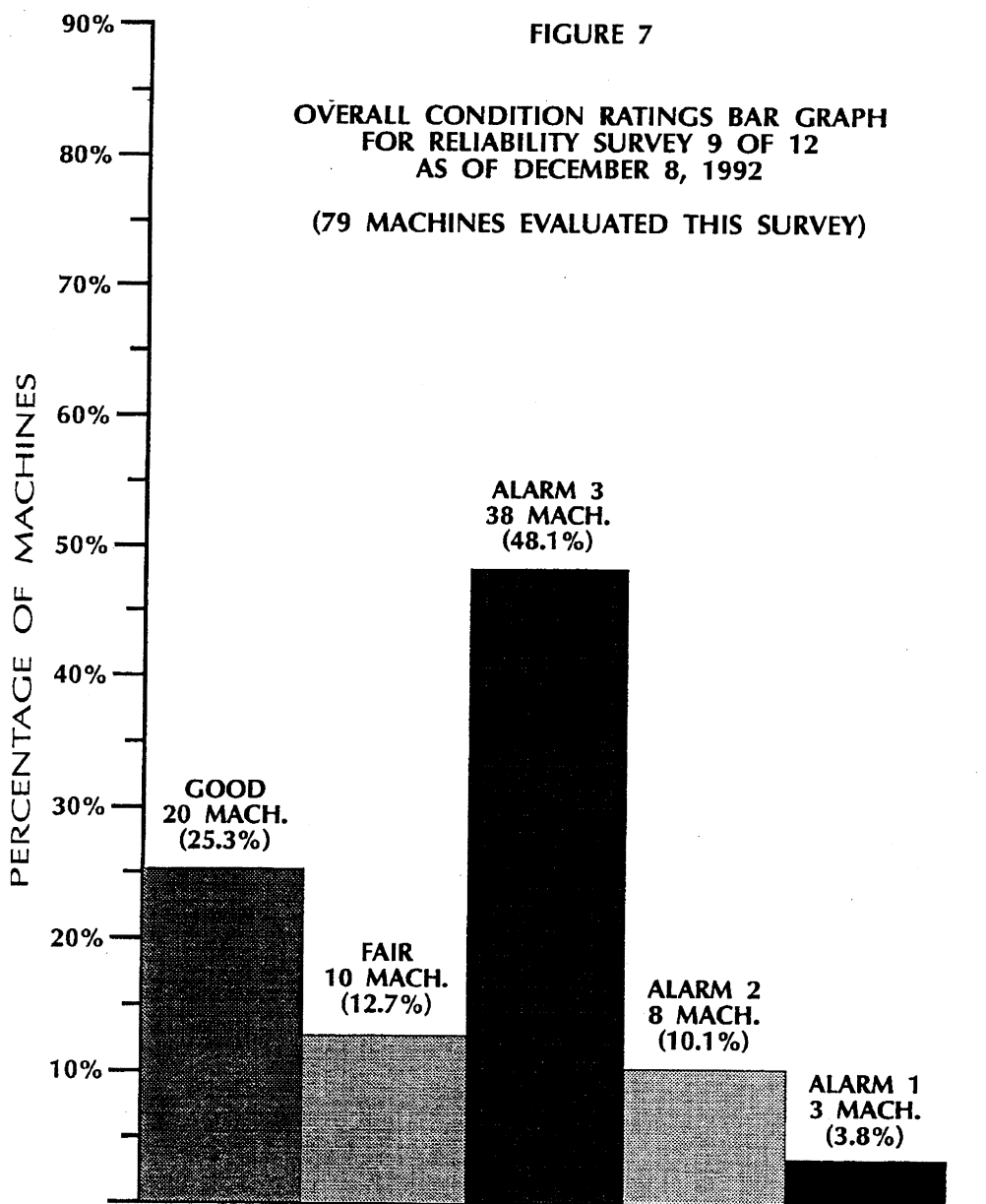
#3 CHILR SUP P5  
#1 VS YORK CHIL  
#4 CHILR SUP P8  
#4 YOR CHILLER  
#5 CHILR SUP P9  
#5 YORK CHILLER

103A COMPRM GEN  
COOL BLWER 101A  
HEATREAT VPMP2  
HOT WTR SUP P-2  
INCIN BCKUP FAN  
INCIN HYD PMP 2

**FIGURE 6**  
**OVERALL CONDITION RATING REPORT**

### FIGURE 7 DESCRIPTION OVERALL CONDITION RATINGS BAR GRAPH

This chart provides a quick glance for PMP personnel and managers into the current state of the plant's machinery condition without getting bogged down in the details of the machine problem or which machines are in alarm. It is useful to trend this information from survey to survey and note if the number of GOOD and FAIR machines grows while the number of machines in ALARM reduces. This provides a useful tool for PMP justification or for future program enhancements.



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## FIGURE 8 DESCRIPTION

### RANK ORDERED SUMMARY OF RESULTS AND RECOMMENDATIONS FOR PROBLEM MACHINES

Problems detected during the **Reliability Survey** are summarized in this section along with Recommendations required to rectify them. Importantly, these are **RANK ORDERED** beginning with the machines considered to have the most severe problems, down to the machines having less serious problems. Not only is each problem identified, but also a priority number (from 1 to 4 with 1 being of highest severity) is used to assess the problem severity and when action should be taken, if any. Also listed is the highest overall vibration position and the measurement at that position for each machine.

**Remarks and Recommendations** are provided on the right-hand-side explaining what is required to successfully repair this machine. The last column lists the plot number(s) for spectra supporting the problem diagnosis.

It is of great importance that the plant follow these recommendations in order that it might achieve fullest possible returns from its **PREDICTIVE MAINTENANCE PROGRAM**. In general, it is recommended that the machines be repaired in the order listed beginning with the #1 problem priority machines. These machines have such high levels of vibrations that they may suffer catastrophic failure if action is not taken quickly. #2 problem priority machines should next be acted upon. Although these machines also have rather significant vibration levels or noticeable component wear problems, it is felt that the Maintenance Department can simply schedule these at their convenience. #3 problem machines require only minor inspection/corrective/diagnostic action such as inspect for loose or broken fasteners, lubricate bearing, or obtain phase data. Note that no action is required by Maintenance on #4 problem priority machines where problems detected on these machines require only trending in future **Reliability Surveys** looking for possible condition deterioration.

In general, it is recommended that neither scheduled nor unscheduled repairs be performed on machinery indicated to be in good operating condition by the PMP program. This will tend to waste expensive maintenance dollars and reduce efficiency both of the program and of the Maintenance staff. However, normal repairs such as periodic lubrication, filter changes, cleaning, fastener tightening, etc. should be continued.

When repairs are made on machines, it is of great importance that the plant notify Technical Associates when the repair was made and what action was taken. This will allow one to evaluate the effectiveness of the repair and will determine whether all problems have been rectified.

**SUMMARY OF RESULTS AND RECOMMENDATIONS FOR MACHINES FOUND TO HAVE PROBLEMS**  
(SORTED FIRST BY PROBLEM PRIORITY; THEN BY PROBLEM SEVERITY WITHIN PRIORITY)

AS OF:

| MACHINE<br>COMPUTER<br>NAME | INDICATED PROBLEM(S) |                        |            |                     |                 |                   |                     |                   |              |                     |            |                  |                |                  | PRIORITY OF PROBLEM | POSITION NO. | LEVEL | HIGHEST OVERALL VIBRATION | RECOMMENDATIONS & REMARKS | SEE PLOT NO. |                    |                |                     |
|-----------------------------|----------------------|------------------------|------------|---------------------|-----------------|-------------------|---------------------|-------------------|--------------|---------------------|------------|------------------|----------------|------------------|---------------------|--------------|-------|---------------------------|---------------------------|--------------|--------------------|----------------|---------------------|
|                             | UNBALANCE            | MISALIGNMENT/SOFT FOOT | BENT SHAFT | LUBRICATION PROBLEM | BEARING PROBLEM | MECHAN. LOOSENESS | BELT/SHEAVE PROBLEM | RESONANCE PROBLEM | GEAR PROBLEM | BLADE PASSING PROB. | CAVITATION | MOTOR ELECTRICAL | OIL WHIRL/WHIP | LOOSE FOUNDATION |                     |              |       |                           |                           |              | VIB. MONITOR PROB. | SEAL MOTOR RUB | OTHER (SEE REMARKS) |
|                             |                      |                        |            |                     |                 |                   |                     |                   |              |                     |            |                  |                |                  |                     |              |       |                           |                           |              |                    |                |                     |
|                             |                      |                        |            |                     |                 |                   |                     |                   |              |                     |            |                  |                |                  |                     |              |       |                           |                           |              |                    |                |                     |

|                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                 |                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 102B SUPY AH-44  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1A<br>2A<br>3BF | 1.741<br>1.511<br>.7401        | The repair log states vibration pads and bracing for the blower were installed. However, high fan speed continues to be measured on the motor due to fan sheave irregularities and likely a weak or resonant motor mounting system. Both motor bearings have indications of rolling element bearing race defects. Inboard fan bearing has peaks which may be belt frequency due to loose belts or the inboard fan bearing turning in its housing or loose on the shaft. Recommend inspecting both fan bearings for proper fits and wear. The motor foundation and fan inboard bearing support obviously need structural stiffening due to weak or resonant structure or loose fasteners. Verify sheaves do not have any excess run-out and replace as needed. Preparations to replace motor bearings should be made and replaced at next scheduled convenience (Priority 2 for motor bearings). Since excessive fan speed vibrations on the motor are likely the cause for the early deterioration of the motor bearings, the fan sheave run-out and fan/motor foundation looseness problems should be corrected before the motor bearings are replaced and as a Priority 1 repair ranking. | 1<br>2<br>3 |
| COND VTR VTRP25  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 2A<br>1A<br>3M  | .5371<br>.2881<br>2.63<br>8/SE | Plots 4 and 5 show harmonics of the inner race defect frequency of a SKF 7313 bearing which is used in this pump as the thrust bearing (T2). Recommend replacing both the up thrust and thrust bearings. As per Ralph Buskirk this will be done during the Christmas 1992 shutdown.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4<br>5      |
| CAFETER AIRU-308 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1V<br>2BF       | .7861<br>.1711                 | Moderate 1X motor RPM noted at motor outboard vertical position likely due to loose/weak motor mounting system. Also, Plot 7 shows possible belt pass frequency and harmonics at Position 2 horizontal likely due to loose, worn, or mismatched belts. The sudden increase at the outboard motor position (Position 1V) may be a result of a loosened mounting bolt or broken structure. Inspect for above and make any corrections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 6<br>7      |

TAKE IMMEDIATE REPAIR ACTION DUE TO SEVERE PROBLEMS NOTED (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)  
SCHEDULE REPAIRS TO CORRECT PROBLEMS NOTED AT CONVENIENCE (SEE "RECOMMENDATIONS/REMARKS" FOR THIS MACHINE)  
DESIGNATED PROBLEM DOES NOT WARRANT ACTION AT THIS TIME. (CONTINUE ONLY WITH RELIABILITY SURVEY TREND ANALYSIS)



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**FIGURE 8**

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## **FIGURE 9 DESCRIPTION PMP MACHINE REPAIR LOG**

This Repair Log is to be completed by the mechanics who make any changes to the machinery being monitored in a PMP program. It is essential that all maintenance actions be recorded on this sheet regardless of whether the repair was ordered by the PMP department or the maintenance department performed alterations to a machine based on orders from themselves or production personnel. The reason this report must be filled out for any machine change is that the vibration signature will change and possibly require a new baselining on which to compare all future vibration measurements. A PMP Machine Repair Log is very crucial to the PMP program as it provides information used in the following manners:

- 1) It tells the PMP personnel that changes have taken place on the machine and a new baseline should possibly be performed.
- 2) The PMP personnel should take a close look at the next set of data collected to determine if other problems have occurred due to the maintenance action taken.
- 3) It is crucial information to monitoring the effectiveness of a PMP program as a repaired machine should show lower vibration levels as well as reduced future maintenance costs.
- 4) It is also a good check to see that the proper maintenance activities were performed and that there was no confusion in the process of getting a repair order from the PMP personnel to the maintenance field personnel.

[illegible]

### Technical Associates Level I

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## FIGURE 10 DESCRIPTION LATEST MEASUREMENTS REPORT

This section contains a listing of all measurement points on the entire machine route. The printout for this section includes:

| <u>COLUMN</u> | <u>TEXT</u> | <u>DESCRIPTION</u>                                                                                 |
|---------------|-------------|----------------------------------------------------------------------------------------------------|
| 1             | (Date)      | Date of Vibration Survey                                                                           |
| 2             | (Point ID)  | Measurement Position and Direction of Transducer<br>(horiz, vert, or axial)                        |
| 3             | (Curr.)     | Overall Amplitude (vibration, spike energy, temperature,<br>RPM, etc.)                             |
| 4             | (Prev.)     | Previous Overall Amplitude (vibration, spike energy,<br>temperature, RPM, etc.)                    |
| 5             | (Units)     | Units of Measurement                                                                               |
| 6             | (% Change)  | Percent Change Since Last Survey *                                                                 |
| 7             | (Warning)   | Preset Overall Alarm Levels (Warning)                                                              |
| 8             | (Critical)  | Preset Overall Alarm Levels (Critical)                                                             |
| 9             | (%Alm)      | Percent of Alarm                                                                                   |
| 10            | (Ave. OA)   | Average Overall Amplitude Based on the Trend of<br>Previously Collected Overall Data (See Fig. 29) |
| 11            | (Std Dv)    | Standard Deviation from Average Overall Amplitude<br>(See Fig. 29)                                 |
| 12            | (PK Freq.)  | Frequency with the Highest Amplitude in Spectrum                                                   |
| 13            | (PK Amp.)   | Amplitude of the Peak Frequency                                                                    |

\* This column reflects the overall vibration % change since the last survey reflecting increases in machinery mechanical deterioration (+) or reductions which may be credited to machinery problem correction (-).

## LATEST OVERALL MEAS. REPORT

| DATE                      | POINT ID       | CURR. PREV. UNITS | %CHG | WARN. CRIT. | %ALM | AVE.OA    | STD DV  | PK FREQ | PK AMPL |
|---------------------------|----------------|-------------------|------|-------------|------|-----------|---------|---------|---------|
| MACHINE : 102B SUPY AH-44 |                |                   |      |             |      |           |         |         |         |
| 12/08/92                  | POS RPM - O    | 1091 1096 RPM     | 0    | 1200 2400   | 91   | 1095.7778 | 14.1755 | No Spec | No Spec |
| 12/08/92                  | POS 1 - A      | 1.740 1.320 IN/S  | 32   | 0.425 0.850 | 409  | 1.6076    | 0.5569  | 1080.00 | 1.73312 |
| 12/08/92                  | POS 1-H HI BPF | 0.303 0.374 g/SE  | -19  | 0.700 1.400 | 43   | 0.4638    | 0.1418  | 6300.00 | 0.09309 |
| 12/08/92                  | POS 1 - H      | 0.272 0.225 IN/S  | 21   | 0.425 0.850 | 64   | 0.2731    | 0.0662  | 1080.00 | 0.26942 |
| 12/08/92                  | POS 1 - V      | 0.950 0.582 IN/S  | 63   | 0.425 0.850 | 224  | 0.9546    | 0.4798  | 1080.00 | 1.30611 |
| 12/08/92                  | POS 2 - A      | 1.510 1.290 IN/S  | 17   | 0.425 0.850 | 355  | 1.2303    | 0.6938  | 1080.00 | 1.38459 |
| 12/08/92                  | POS 2-H HI BPF | 0.248 0.251 g/SE  | -1   | 0.700 1.400 | 35   | 0.3871    | 0.2086  | 6300.00 | 0.05600 |
| 12/08/92                  | POS 2 - O      | 0.235 0.215 IN/S  | 9    | 0.425 0.850 | 55   | 0.2993    | 0.0731  | 1170.00 | 0.14035 |
| 12/08/92                  | POS 2BF        | 0.264 0.262 IN/S  | 1    | 0.425 0.850 | 62   | 0.2947    | 0.0649  | 1170.00 | 0.18888 |
| 12/08/92                  | POS 2 - H      | 0.314 0.266 IN/S  | 18   | 0.425 0.850 | 74   | 0.3063    | 0.0763  | 1080.00 | 0.19722 |
| 12/08/92                  | POS 2 - V      | 0.710 0.870 IN/S  | -18  | 0.425 0.850 | 167  | 0.6853    | 0.1626  | 1800.00 | 0.54504 |
| 12/08/92                  | POS 3-H HI BPF | 0.432 0.482 g/SE  | -10  | 0.450 0.900 | 96   | 0.4122    | 0.1261  | 150.00  | 0.62716 |
| 12/08/92                  | POS 3 - H      | 0.594 0.480 IN/S  | 24   | 0.425 0.850 | 140  | 0.7110    | 0.2261  | 1080.00 | 0.67434 |
| 12/08/92                  | POS 3BF        | 0.740 0.410 IN/S  | 80   | 0.425 0.850 | 174  | 0.6103    | 0.1388  | 1095.00 | 0.55536 |
| 12/08/92                  | POS 3 - V      | 0.305 0.735 IN/S  | -59  | 0.425 0.850 | 72   | 0.8143    | 0.2915  | 1080.00 | 0.14932 |
| MACHINE : CAFETER AHU-308 |                |                   |      |             |      |           |         |         |         |
| 12/08/92                  | POS RPM - O    | 1255 1257 RPM     | 0    | 1400 2800   | 90   | 1266.5556 | 8.5602  | No Spec | No Spec |
| 12/08/92                  | POS 1 - A      | 0.052 0.039 g/SE  | 33   | 0.700 1.400 | 7    | 0.0402    | 0.0178  | No Spec | No Spec |
| 12/08/92                  | POS 1 - A      | 0.222 0.104 IN/S  | 113  | 0.425 0.850 | 52   | 0.1994    | 0.0898  | 1800.00 | 0.20278 |
| 12/08/92                  | POS 1 - H      | 0.315 0.260 IN/S  | 21   | 0.425 0.850 | 74   | 0.3155    | 0.0676  | 1800.00 | 0.24694 |
| 12/08/92                  | POS 1 - V      | 0.786 0.476 IN/S  | 65   | 0.425 0.850 | 185  | 0.6031    | 0.1792  | 1800.00 | 0.71374 |
| 12/08/92                  | POS 2 - A      | 0.035 0.017 g/SE  | 106  | 0.700 1.400 | 5    | 0.0305    | 0.0158  | No Spec | No Spec |
| 12/08/92                  | POS 2 - A      | 0.182 0.365 IN/S  | -50  | 0.425 0.850 | 43   | 0.2372    | 0.0873  | 1800.00 | 0.15695 |
| 12/08/92                  | POS 2 - O      | 0.160 0.185 IN/S  | -14  | 0.425 0.850 | 38   | 0.2090    | 0.0564  | 1260.00 | 0.09801 |
| 12/08/92                  | POS 2BF        | 0.171 0.189 IN/S  | -10  | 0.425 0.850 | 40   | 0.1803    | 0.0113  | 315.00  | 0.09665 |
| 12/08/92                  | POS 2 - H      | 0.151 0.205 IN/S  | -26  | 0.425 0.850 | 36   | 0.2109    | 0.0650  | 2520.00 | 0.09102 |
| 12/08/92                  | POS 2 - V      | 0.165 0.229 IN/S  | -28  | 0.425 0.850 | 39   | 0.2134    | 0.0476  | 1800.00 | 0.11774 |
| 12/08/92                  | POS 3 - A      | 0.064 0.060 g/SE  | 7    | 0.500 1.000 | 13   | 0.0941    | 0.0775  | No Spec | No Spec |
| 12/08/92                  | POS 3 - A      | 0.247 0.198 IN/S  | 25   | 0.425 0.850 | 58   | 0.2045    | 0.0455  | 2550.00 | 0.16704 |
| 12/08/92                  | POS 3 - H      | 0.162 0.184 IN/S  | -12  | 0.425 0.850 | 38   | 0.1775    | 0.0431  | 1200.00 | 0.09385 |
| 12/08/92                  | POS 3BF        | 0.139 0.178 IN/S  | -22  | 0.425 0.850 | 33   | 0.1598    | 0.0173  | 1260.00 | 0.10790 |
| 12/08/92                  | POS 3 - V      | 0.166 0.165 IN/S  | 1    | 0.425 0.850 | 39   | 0.1647    | 0.0187  | 1200.00 | 0.09484 |
| 12/08/92                  | POS 4 - A      | 0.076 0.105 g/SE  | -28  | 0.500 1.000 | 15   | 0.0976    | 0.0485  | No Spec | No Spec |
| 12/08/92                  | POS 4 - A      | 0.193 0.131 IN/S  | 47   | 0.425 0.850 | 45   | 0.2063    | 0.0543  | 1800.00 | 0.14273 |
| 12/08/92                  | POS 4 - H      | 0.185 0.131 IN/S  | 41   | 0.425 0.850 | 44   | 0.2304    | 0.0907  | 1200.00 | 0.11171 |
| 12/08/92                  | POS 4 - V      | 0.141 0.112 IN/S  | 26   | 0.425 0.850 | 33   | 0.1513    | 0.0336  | 1800.00 | 0.08060 |
| MACHINE : COND WTR VPMP25 |                |                   |      |             |      |           |         |         |         |
| 12/08/92                  | POS 1A - A     | 0.288 0.178 IN/S  | 62   | 0.200 0.400 | 144  | 0.0942    | 0.0559  | 25350.0 | 0.14861 |
| 12/08/92                  | POS 1W - O     | 0.163 0.098 IN/S  | 66   | 0.350 0.700 | 47   | 0.1182    | 0.0677  | 1800.00 | 0.03711 |
| 12/08/92                  | POS 1N - O     | 0.136 0.182 IN/S  | -25  | 0.350 0.700 | 39   | 0.1215    | 0.0610  | 1800.00 | 0.10945 |
| 12/08/92                  | POS 1N - O     | 0.409 0.195 g/SE  | 110  | 0.700 1.400 | 58   | 0.1526    | 0.0879  | No Spec | No Spec |
| 12/08/92                  | POS 1EO - O    | 0.146 0.205 IN/S  | -29  | 0.350 0.700 | 42   | 0.1318    | 0.0519  | 78000.0 | 0.06345 |
| 12/08/92                  | POS 2A - A     | 0.211 0.129 IN/S  | 64   | 0.200 0.400 | 106  | 0.0765    | 0.0414  | 12750.0 | 0.07728 |
| 12/08/92                  | POS 2W - O     | 0.164 0.137 IN/S  | 20   | 0.300 0.600 | 55   | 0.1009    | 0.0322  | 25350.0 | 0.03320 |
| 12/08/92                  | POS 2N - O     | 0.226 0.186 IN/S  | 22   | 0.300 0.600 | 75   | 0.1009    | 0.0476  | 49050.0 | 0.06696 |
| 12/08/92                  | POS 2N - O     | 0.327 0.539 g/SE  | -39  | 0.700 1.400 | 47   | 0.1865    | 0.1284  | No Spec | No Spec |
| 12/08/92                  | POS 2EO - O    | 0.251 0.144 IN/S  | 74   | 0.300 0.600 | 84   | 0.1162    | 0.0527  | 1800.00 | 0.19039 |
| 12/08/92                  | POS 3A - A     | 0.537 0.421 IN/S  | 28   | 0.200 0.400 | 269  | 0.1867    | 0.1365  | 49050.0 | 0.34154 |
| 12/08/92                  | POS 3W - O     | 0.433 0.228 IN/S  | 90   | 0.300 0.600 | 144  | 0.1341    | 0.0971  | 49050.0 | 0.04575 |
| 12/08/92                  | POS 3N - O     | 0.333 0.179 IN/S  | 86   | 0.300 0.600 | 111  | 0.1139    | 0.0847  | 39900.0 | 0.04956 |
| 12/08/92                  | POS 3N - O     | 2.630 1.220 g/SE  | 116  | 0.700 1.400 | 376  | 0.8009    | 0.7211  | No Spec | No Spec |

FIGURE 10

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## FIGURE 11 DESCRIPTION ALARM REPORTS

The **OVERALL ALARM EXCEPTION REPORT** detects all of the machines that have vibration which has exceeded the Overall Alarm for one or more points (not necessarily have exceeded the spectral alarms). These machines are often in need of corrective maintenance and are therefore of primary importance. Many of the machines may have vibration well in excess of the predetermined alarm levels on an initial survey and these need particular attention.

On the following pages are computer printouts of the **OVERALL ALARM EXCEPTION REPORT**. These columns are:

| <u>COLUMN</u> | <u>TEXT</u> | <u>DESCRIPTION</u>                                                                                                                 |
|---------------|-------------|------------------------------------------------------------------------------------------------------------------------------------|
| 1             | (Date)      | Measurement Date                                                                                                                   |
| 2             | (Point ID)  | Position Number Where Measurement was Taken and Direction of Transducer - Either Axial (A), Horizontal (H), or Vertical (V)        |
| 3             | (Curr.)     | Overall Amplitude of Current Vibration                                                                                             |
| 4             | (Prev.)     | Overall Amplitude of Previous Vibration                                                                                            |
| 5             | (Units)     | Units of Vibration, whether in Velocity (In/Sec), Displacement (Mils), Acceleration (g), Spike Energy (gSE), Temperature (°F), etc |
| 6             | (Warn.)     | Preset Overall Alarm 1 Levels (Warning)                                                                                            |
| 7             | (Crit.)     | Preset Overall Alarm 2 Levels (Critical)                                                                                           |
| 8             | (%Alm)      | Percentage of Overall Warning Alarm (100% = At Alarm Level)                                                                        |
| 9             | (% Chg)     | Percentage of Change since Previous Measurement                                                                                    |
| 10            | (Ave. OA)   | Average Overall Amplitude Based on the Trend of all Previously Collected Overall Data Stored to Date (See Fig. 29)                 |
| 11            | (Std Dv)    | Standard Deviation from Average Overall Amplitude (See Fig. 29)                                                                    |
| 12            | (PK Freq.)  | Frequency with the Highest Amplitude in the Spectrum for this Point                                                                |
| 13            | (PK Amp.)   | Amplitude of the Peak Frequency (mils, in/sec, g, gSE, etc.)                                                                       |

# OVERALL ALARM EXCEPTION REPORT

| DATE                      | POINT ID | CURR. PREV. UNITS | WARN. CRIT. %ALM %CHG | AVE.OA STD DV PK FREQ PK AMPL |
|---------------------------|----------|-------------------|-----------------------|-------------------------------|
| MACHINE : 102B SUPY AH-44 |          |                   |                       |                               |
| 12/08/92 POS              | 1 - A    | 1.740 1.320 IN/S  | 0.425 0.850 409 32    | 1.6076 0.5569 1080.00 1.73312 |
| 12/08/92 POS              | 1 - V    | 0.950 0.582 IN/S  | 0.425 0.850 224 63    | 0.9546 0.4798 1080.00 1.30611 |
| 12/08/92 POS              | 2 - A    | 1.510 1.290 IN/S  | 0.425 0.850 355 17    | 1.2303 0.6938 1080.00 1.38459 |
| 12/08/92 POS              | 2 - V    | 0.710 0.870 IN/S  | 0.425 0.850 167 -18   | 0.6853 0.1626 1800.00 0.54504 |
| 12/08/92 POS              | 3 - H    | 0.594 0.480 IN/S  | 0.425 0.850 140 24    | 0.7110 0.2261 1080.00 0.67434 |
| 12/08/92 POS              | 38F      | 0.740 0.410 IN/S  | 0.425 0.850 174 80    | 0.6103 0.1388 1095.00 0.55536 |
| MACHINE : CAFETER AHU-308 |          |                   |                       |                               |
| 12/08/92 POS              | 1 - V    | 0.786 0.476 IN/S  | 0.425 0.850 185 65    | 0.6031 0.1792 1800.00 0.71374 |
| MACHINE : COND WTR VPMP25 |          |                   |                       |                               |
| 12/08/92 POS              | 1A - A   | 0.288 0.178 IN/S  | 0.200 0.400 144 62    | 0.0942 0.0559 25350.0 0.14861 |
| 12/08/92 POS              | 2A - A   | 0.211 0.129 IN/S  | 0.200 0.400 106 64    | 0.0765 0.0414 12750.0 0.07728 |
| 12/08/92 POS              | 3A - A   | 0.537 0.421 IN/S  | 0.200 0.400 269 28    | 0.1867 0.1365 49050.0 0.34154 |
| 12/08/92 POS              | 3W - O   | 0.433 0.228 IN/S  | 0.300 0.600 144 90    | 0.1341 0.0971 49050.0 0.04575 |
| 12/08/92 POS              | 3N - O   | 0.333 0.179 IN/S  | 0.300 0.600 111 86    | 0.1139 0.0847 39900.0 0.04956 |
| 12/08/92 POS              | 3N - O   | 2.630 1.220 g/SE  | 0.700 1.400 376 116   | 0.8009 0.7211 No Spec No Spec |

FIGURE 11

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## FIGURE 12 DESCRIPTION SPECTRAL BAND ALARM REPORT

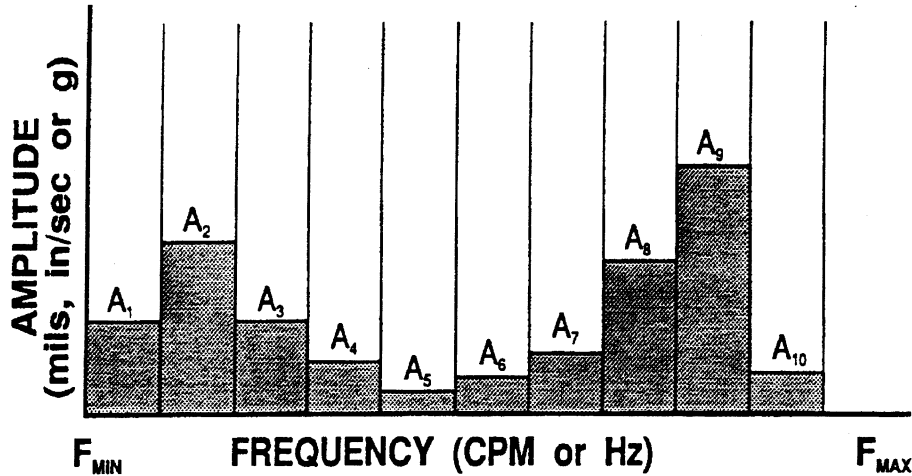
The **SPECTRAL BAND ALARM REPORT** detects all points on all PMP machines in which the vibration alarm level has been exceeded in 1 or more of the 6 spectral bands. The spectral bands have differing alarm levels in each of the 6 individual bands into which the spectrum is divided. It is important to know that these are "Power Bands" - that is, that the vibration within each band equals the RSS (Root Sum Square) level calculated for each band \*. This allow higher vibration levels at 1X RPM which is normally high due to some amount of unbalance usually being present, while allowing much lower levels of vibration near bearing defect frequencies (which should not be present at all). These machines are often in need of corrective maintenance and are therefore of primary importance. Many of the machines may have vibration well in excess of the predetermined alarm levels on an initial survey and these need particular attention.

Importantly, if the RSS vibration within a band exceeds its specified alarm, an entry denoted "YES" will be printed out beside this particular band(s) in this report for each individual point listed. It is important to realize that it is not necessary for any peak within a band to exceed the band alarm - only that the total RSS value for the band equal or exceed this alarm.

Also on the following pages are computer printouts of the **SPECTRAL BAND ALARM REPORT** which has 14 columns. These 14 columns are:

| <u>COLUMN</u> | <u>TEXT</u> | <u>DESCRIPTION</u>                                                                                                     |
|---------------|-------------|------------------------------------------------------------------------------------------------------------------------|
| 1             | (Date)      | Measurement Date                                                                                                       |
| 2             | (Point ID)  | Position Number Where Measurement was Taken and Direction of Probe - Either Axial (A), Horizontal (H), or Vertical (V) |
| 3             | (B1)        | Band 1 Amplitude                                                                                                       |
| 4             | (ALA)       | If Band Alarm is Exceeded This Column says "YES"                                                                       |
| 5             | (B2)        | Band 2 Amplitude                                                                                                       |
| 6             | (ALA)       | If Band Alarm is Exceeded This Column says "YES"                                                                       |
| 7             | (B3)        | Band 3 Amplitude                                                                                                       |
| 8             | (ALA)       | If Band Alarm is Exceeded This Column says "YES"                                                                       |
| 9             | (B4)        | Band 4 Amplitude                                                                                                       |
| 10            | (ALA)       | If Band Alarm is Exceeded This Column says "YES"                                                                       |
| 11            | (B5)        | Band 5 Amplitude                                                                                                       |
| 12            | (ALA)       | If Band Alarm is Exceeded This Column says "YES"                                                                       |
| 13            | (B6)        | Band 6 Amplitude                                                                                                       |
| 14            | (ALA)       | If Band Alarm is Exceeded This Column says "YES"                                                                       |

\*NOTE: "RSS Amplitude" = Root Sum Square Amplitude ( in layman's terms, it is a "summation of energy" within the band, which includes the amplitudes of each FFT line from the beginning of the band to the end of the band). This is illustrated in Illustration 1.



$$OA_{RSS} = \frac{\sqrt{\sum_{i=1}^n A_i^2}}{\sqrt{N_{BF}}} = \frac{\sqrt{A_1^2 + A_2^2 + A_3^2 + \dots + A_n^2}}{\sqrt{1.5}}$$

**$OA_{RSS}$  = Digital Overall Level of Vibration Spectrum.**

**$RSS$  = Root Sum Square Level which is the "Summation of Energy" in the spectrum from  $F_{MIN}$  to  $F_{MAX}$ .**

**$n$  = Number of FFT Lines of Resolution.**

**$A_i$  = Amplitude of Each of the FFT Lines.**

**$N_{BF}$  = Noise Bandwidth for Window Chosen 1.5 for Hanning Window.**

Which means  $OA_{RSS} = .8165 \sqrt{\sum_{i=1}^n A_i^2}$

**ILLUSTRATION 1  
EXACT EQUATION FOR CALCULATING DIGITAL  
OVERALL LEVEL OF A SPECTRUM**

# BAND ALARM REPORT

| DATE                      | POINT ID    | B1 ALM     | B2 ALM     | B3 ALM     | B4 ALM     | B5 ALM     | B6 ALM     |
|---------------------------|-------------|------------|------------|------------|------------|------------|------------|
| MACHINE : 102B SUPY AH-44 |             |            |            |            |            |            |            |
| 12/08/92                  | POS 1 - A   | 1.8260 YES | 0.0768 --- | 0.0123 --- | 0.1064 --- | 0.0159 --- | 0.0570 --- |
| 12/08/92                  | POS 1 - V   | 1.5542 YES | 0.1053 --- | 0.0060 --- | 0.0152 --- | 0.0107 --- | 0.0182 --- |
| 12/08/92                  | POS 2 - A   | 1.4431 YES | 0.0794 --- | 0.0137 --- | 0.0824 --- | 0.0118 --- | 0.0362 --- |
| 12/08/92                  | POS 2BF     | 0.0115 --- | 0.1838 --- | 0.1449 YES | 0.0773 --- | 0.0376 --- | 0.0328 --- |
| 12/08/92                  | POS 2 - V   | 0.6363 YES | 0.0778 --- | 0.0235 --- | 0.0436 --- | 0.0207 --- | 0.0345 --- |
| 12/08/92                  | POS 3 - H   | 0.7700 YES | 0.1247 --- | 0.0619 --- | 0.0629 --- | 0.0167 --- | 0.0307 --- |
| 12/08/92                  | POS 3BF     | 0.1385 YES | 0.6765 YES | 0.3511 YES | 0.0887 --- | 0.0530 --- | 0.0864 YES |
| MACHINE : CAFETER AHU-308 |             |            |            |            |            |            |            |
| 12/08/92                  | POS 1 - V   | 0.7380 YES | 0.0922 --- | 0.0731 --- | 0.0101 --- | 0.0046 --- | 0.0029 --- |
| 12/08/92                  | POS 2BF     | 0.1035 YES | 0.1066 --- | 0.0085 --- | 0.0907 --- | 0.0100 --- | 0.0268 --- |
| MACHINE : COND WTR VPMP25 |             |            |            |            |            |            |            |
| 12/08/92                  | POS 1A - A  | 0.0776 --- | 0.0060 --- | 0.0110 --- | 0.1055 YES | 0.1832 YES | 0.0494 YES |
| 12/08/92                  | POS 1EO - O | 0.0458 --- | 0.1099 YES | 0.0194 --- | 0.0059 --- | 0.0034 --- | 0.0014 --- |
| 12/08/92                  | POS 2A - A  | 0.0323 --- | 0.0106 --- | 0.0075 --- | 0.1042 YES | 0.0742 YES | 0.0361 --- |
| 12/08/92                  | POS 2W - O  | 0.0522 --- | 0.0067 --- | 0.0039 --- | 0.0422 --- | 0.0583 --- | 0.0734 YES |
| 12/08/92                  | POS 3A - A  | 0.0128 --- | 0.0078 --- | 0.0107 --- | 0.0367 --- | 0.0902 YES | 0.5254 YES |

FIGURE 12

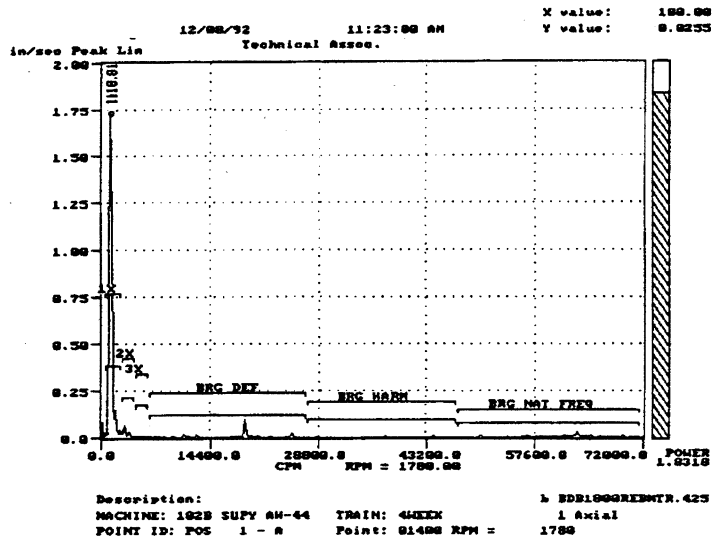
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## FIGURES 13 THRU 19 DESCRIPTION GRAPHICS REPORT FOR SPECTRA HAVING SPECTRAL ALARM BANDS

### A. Vibration Spectra

This section includes plots of Amplitude versus Frequency (with Spectral Alarm Bands displayed on them) which were referenced in "**RANK-ORDERED RESULTS AND RECOMMENDATIONS REPORT**" (Figure 8). Spectra are included for those machines which have either experienced significant vibration increases, are now in **ALARM**, or which have individual frequencies indicating duress of a particular machine component(s) whether or not the amplitude of these particular frequencies have increased overall levels into **ALARM** severity. Where applicable, additional notation is provided on the plot relating information on problem components and/or conditions which have been detected. Machine name, point location, vibration parameter, amplitude scale and frequency range are provided on each plot, along with spectral band alarms.

Figures 13 thru 19 (Plots 1 thru 7) are the plots of the spectra from the **RANK-ORDERED SUMMARY OF RESULTS AND RECOMMENDATIONS FOR PROBLEM MACHINES** (See Figure 8) that have been diagnosed as having problems. Note that the "Plot Nos." have been referred to in the last column of this report so that the reader can compare the diagnosed problem to the spectra from which the diagnosis was made.



12/08/92  
 11:23:00 AM

### Threshold Report

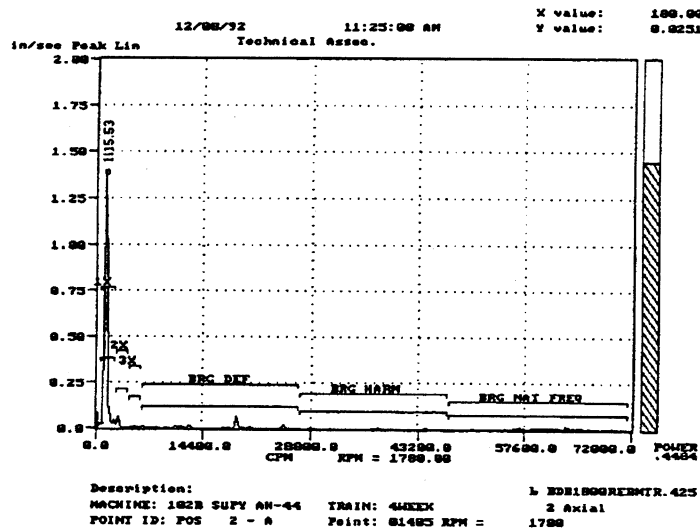
PLANT:  
 Description:  
 MACHINE: 102B SUPY AH-44  
 POINT ID: POS 1 - A  
 Units: in/sec Peak Lin  
 Threshold: 0.0100

TRAIN: 4WEEK  
 Point: 01400  
 RPM: 1780.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 3.00         | 180     | 0.10    | 0.0255    | 154.53     |
| 18.65        | 1.12E+3 | 0.63    | 1.7331    | 17231.20   |
| 48.97        | 2938    | 1.65    | 0.0629    | 528.84     |
| 58.62        | 3517    | 1.98    | 0.0279    | 178.62     |
| 180.98       | 1.09E+4 | 6.10    | 0.0200    | 99.98      |
| 210.00       | 12600   | 7.08    | 0.0157    | 57.12      |
| 314.91       | 1.89E+4 | 10.62   | 0.0959    | 859.07     |
| 330.57       | 19834   | 11.14   | 0.0103    | 3.02       |
| 343.96       | 2.06E+4 | 11.59   | 0.0103    | 3.02       |
| 419.91       | 2.52E+4 | 14.15   | 0.0247    | 146.97     |
| 944.56       | 56673   | 31.84   | 0.0103    | 3.02       |
| 1038.36      | 6.23E+4 | 35.00   | 0.0120    | 19.78      |
| 1056.77      | 6.34E+4 | 35.62   | 0.0305    | 2.05E+2    |
| 1074.95      | 64497   | 36.23   | 0.0103    | 3.02       |
| 1155.23      | 6.93E+4 | 38.94   | 0.0103    | 3.02       |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 1.8318       | 0.1592  | 1.8194  | 0.1404    |            |

PLOT 1

FIGURE 13



12/08/92  
11:25:00 AM

### Threshold Report

PLANT:  
Description:

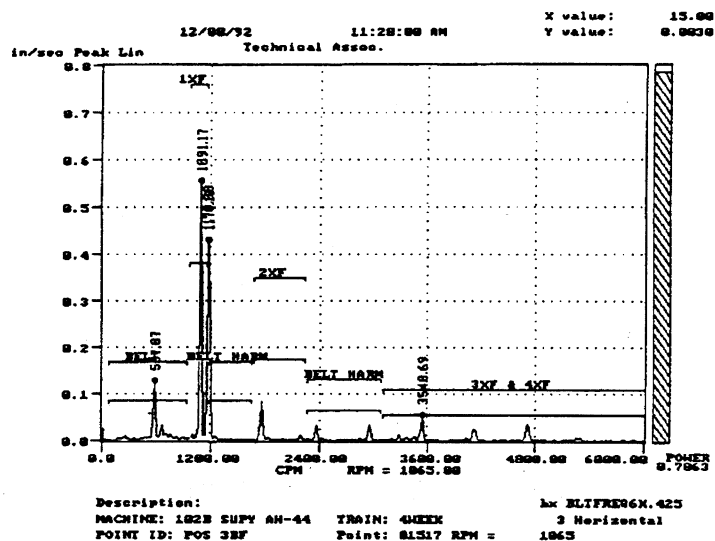
MACHINE: 102B SUPY AH-44  
POINT ID: POS 2 - A  
Units: in/sec Peak Lin  
Threshold: 0.0100

TRAIN: 4WEEK  
Point: 01405  
RPM: 1780.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 3.00         | 180     | 0.10    | 0.0251    | 151.29     |
| 18.59        | 1.12E+3 | 0.63    | 1.3846    | 13745.90   |
| 39.32        | 2359    | 1.33    | 0.0485    | 384.85     |
| 49.24        | 2954    | 1.66    | 0.0673    | 573.48     |
| 105.91       | 6.35E+3 | 3.57    | 0.0130    | 30.24      |
| 181.16       | 10869   | 6.11    | 0.0122    | 21.95      |
| 190.60       | 11436   | 6.42    | 0.0107    | 6.93       |
| 210.70       | 1.26E+4 | 7.10    | 0.0206    | 106.32     |
| 315.94       | 18956   | 10.65   | 0.0652    | 551.71     |
| 421.25       | 25275   | 14.20   | 0.0228    | 127.70     |
| 947.42       | 56845   | 31.94   | 0.0122    | 21.95      |
| 1052.74      | 6.32E+4 | 35.49   | 0.0153    | 53.50      |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 1.4484       | 0.1202  | 1.4380  | 0.1241    |            |

PLOT 2

FIGURE 14



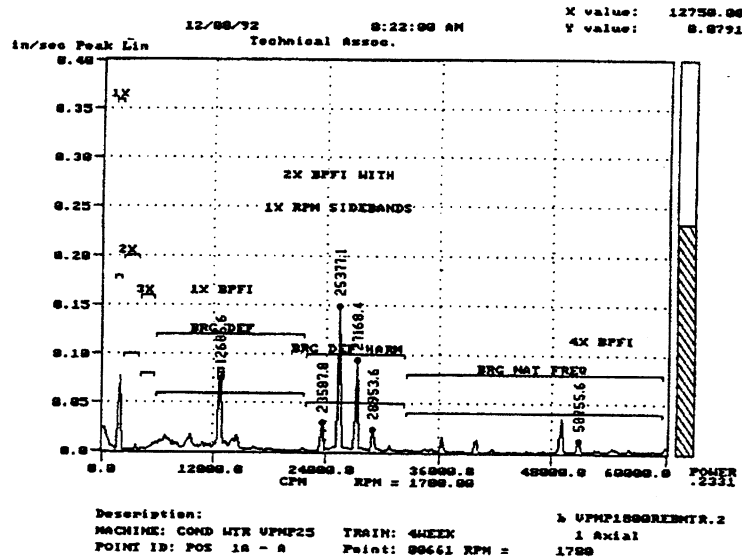
12/08/92  
11:28:00 AM

### Threshold Report

PLANT:  
Description:  
MACHINE: 102B SUPY AH-44 TRAIN: 4WEEK  
POINT ID: POS 3BF Point: 01517  
Units: in/sec Peak Lin RPM: 1065.00  
Threshold: 0.0100

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 4.20         | 2.52E+2 | 0.24    | 0.0112    | 12.00      |
| 9.80         | 5.88E+2 | 0.55    | 0.1275    | 1174.51    |
| 11.17        | 670     | 0.63    | 0.0322    | 221.96     |
| 11.92        | 7.15E+2 | 0.67    | 0.0149    | 49.38      |
| 12.77        | 766     | 0.72    | 0.0136    | 35.71      |
| 16.72        | 1.00E+3 | 0.94    | 0.0149    | 49.38      |
| 18.19        | 1091    | 1.02    | 0.5554    | 5453.61    |
| 19.62        | 1177    | 1.11    | 0.4299    | 4199.37    |
| 20.85        | 1251    | 1.17    | 0.0102    | 1.75       |
| 29.27        | 1756    | 1.65    | 0.0841    | 740.80     |
| 36.39        | 2183    | 2.05    | 0.0108    | 8.47       |
| 39.31        | 2.36E+3 | 2.21    | 0.0332    | 232.43     |
| 49.15        | 2949    | 2.77    | 0.0332    | 232.43     |
| 54.59        | 3.28E+3 | 3.08    | 0.0140    | 40.12      |
| 59.01        | 3.54E+3 | 3.32    | 0.0555    | 454.67     |
| 68.65        | 4.12E+3 | 3.87    | 0.0266    | 165.72     |
| 78.47        | 4708    | 4.42    | 0.0366    | 265.92     |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.7863       | 0.3101  | 0.1394  | 0.7090    |            |

PLOT 3  
FIGURE 15



12/08/92  
8:22:00 AM

### Threshold Report

PLANT:  
Description:

MACHINE: COND WTR VPMP25  
POINT ID: POS 1A - A

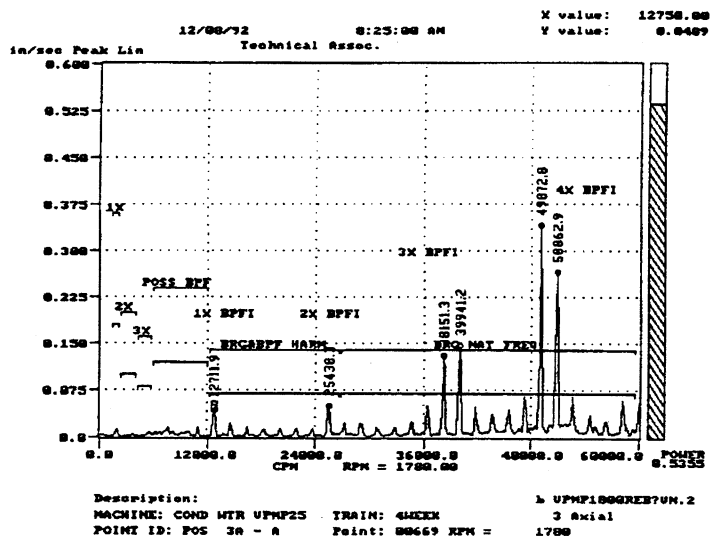
Units: in/sec Peak Lin  
Threshold: 0.0100

TRAIN: 4WEEK  
Point: 00661  
RPM: 1780.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 2.50         | 150     | 0.08    | 0.0241    | 141.48     |
| 29.88        | 1.79E+3 | 1.01    | 0.0771    | 671.03     |
| 103.65       | 6219    | 3.49    | 0.0105    | 5.00E+0    |
| 113.56       | 6.81E+3 | 3.83    | 0.0165    | 65.38      |
| 127.50       | 7650    | 4.30    | 0.0102    | 2.38       |
| 158.53       | 9.51E+3 | 5.34    | 0.0170    | 69.60      |
| 211.44       | 1.27E+4 | 7.13    | 0.0791    | 690.74     |
| 241.48       | 1.45E+4 | 8.14    | 0.0170    | 69.60      |
| 393.13       | 2.36E+4 | 13.25   | 0.0296    | 195.51     |
| 422.95       | 25377   | 14.26   | 0.1486    | 1386.10    |
| 452.81       | 27168   | 15.26   | 0.0944    | 843.53     |
| 482.56       | 2.90E+4 | 16.27   | 0.0224    | 123.87     |
| 604.34       | 36260   | 20.37   | 0.0153    | 53.32      |
| 664.11       | 3.98E+4 | 22.39   | 0.0128    | 28.49      |
| 815.98       | 4.90E+4 | 27.50   | 0.0362    | 261.62     |
| 845.93       | 5.08E+4 | 28.51   | 0.0119    | 19.12      |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.2332       | 0.1025  | 0.0321  | 0.2069    |            |

PLOT 4

FIGURE 16



12/08/92  
 8:25:00 AM

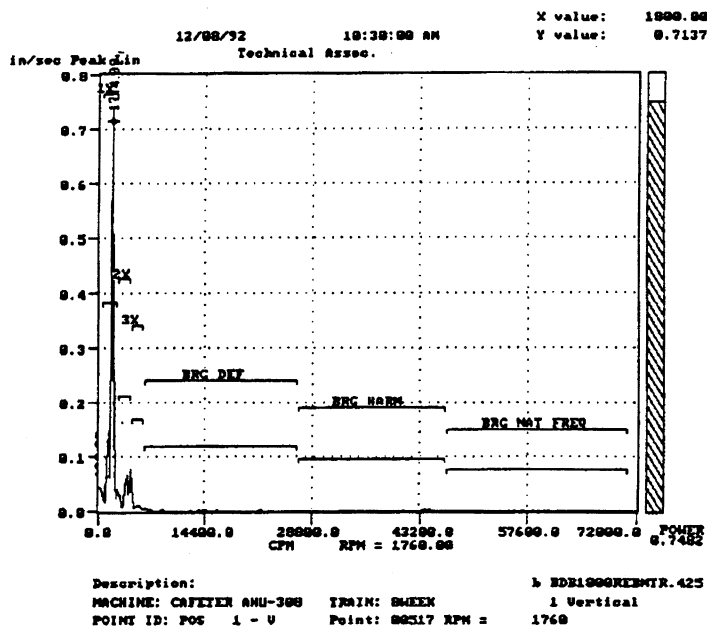
### Threshold Report

PLANT:  
 Description:  
 MACHINE: COND WTR VPMP25 TRAIN: 4WEEK  
 POINT ID: POS 3A - A Point: 00669  
 Units: in/sec Peak Lin RPM: 1780.00  
 Threshold: 0.0200

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 211.87       | 1.27E+4 | 7.14    | 0.0489    | 144.55     |
| 241.73       | 1.45E+4 | 8.15    | 0.0234    | 16.79      |
| 423.84       | 25430   | 14.29   | 0.0503    | 151.34     |
| 453.65       | 2.72E+4 | 15.29   | 0.0240    | 20.03      |
| 483.66       | 29019   | 16.30   | 0.0234    | 16.79      |
| 575.98       | 3.46E+4 | 19.42   | 0.0254    | 26.78      |
| 605.85       | 36351   | 20.42   | 0.0503    | 151.34     |
| 635.86       | 38151   | 21.43   | 0.1310    | 555.14     |
| 665.69       | 39941   | 22.44   | 0.1462    | 630.95     |
| 695.64       | 4.17E+4 | 23.45   | 0.0476    | 137.95     |
| 726.37       | 43582   | 24.48   | 0.0372    | 85.99      |
| 757.37       | 45442   | 25.53   | 0.0438    | 119.19     |
| 787.90       | 4.73E+4 | 26.56   | 0.0661    | 230.48     |
| 807.36       | 48441   | 27.21   | 0.0204    | 1.85       |
| 817.88       | 4.91E+4 | 27.57   | 0.3415    | 1607.70    |
| 835.82       | 50149   | 28.17   | 0.0209    | 4.68       |
| 847.72       | 5.09E+4 | 28.57   | 0.2670    | 1234.82    |
| 864.32       | 5.19E+4 | 29.13   | 0.0240    | 20.03      |
| 877.88       | 5.27E+4 | 29.59   | 0.0661    | 230.48     |
| 909.29       | 5.46E+4 | 30.65   | 0.0362    | 80.97      |
| 938.67       | 56320   | 31.64   | 0.0254    | 26.78      |
| 969.96       | 5.82E+4 | 32.70   | 0.0592    | 196.20     |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.5373       | 0.0711  | 0.0056  | 0.5326    |            |

PLOT 5

FIGURE 17



12/08/92  
 10:30:00 AM

### Threshold Report

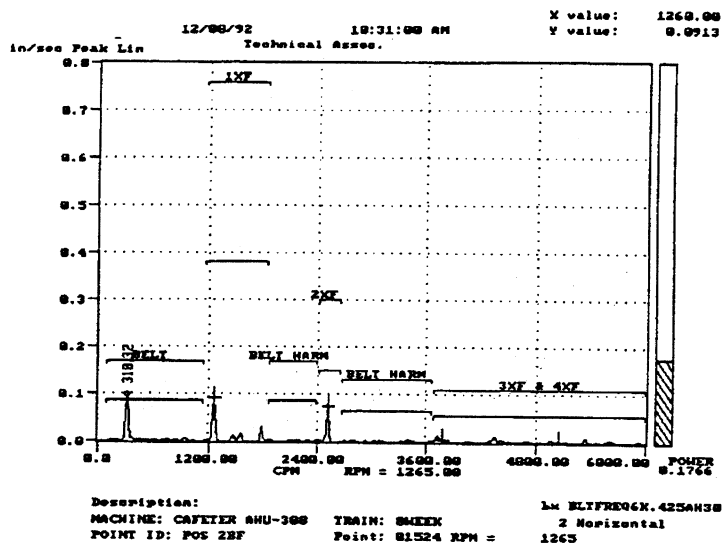
PLANT:  
 Description:  
 MACHINE: CAFETER AHU-308  
 POINT ID: POS 1 - V  
 Units: in/sec Peak Lin  
 Threshold: 0.0100

TRAIN: 8WEEK  
 Point: 00517  
 RPM: 1760.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 3.00         | 180     | 0.10    | 0.0419    | 318.51     |
| 29.57        | 1774    | 1.01    | 0.7137    | 7037.40    |
| 62.27        | 3736    | 2.12    | 0.0664    | 564.10     |
| 72.21        | 4.33E+3 | 2.46    | 0.0758    | 657.76     |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.7482       | 0.7237  | 0.1579  | 0.1058    |            |

PLOT 6

FIGURE 18



12/08/92  
 10:31:00 AM

### Threshold Report

PLANT:  
 Description:  
 MACHINE: CAFETER AHU-308  
 POINT ID: POS 2BF  
 Units: in/sec Peak Lin  
 Threshold: 0.0100

TRAIN: 8WEEK  
 Point: 01524  
 RPM: 1265.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 5.17         | 310     | 0.25    | 0.0966    | 866.48     |
| 20.91        | 1.25E+3 | 0.99    | 0.0913    | 812.51     |
| 24.43        | 1.47E+3 | 1.16    | 0.0112    | 12.01      |
| 25.81        | 1.55E+3 | 1.22    | 0.0167    | 67.49      |
| 29.64        | 1.78E+3 | 1.41    | 0.0315    | 215.15     |
| 41.87        | 2.51E+3 | 1.99    | 0.0790    | 690.39     |
| 61.94        | 3716    | 2.94    | 0.0137    | 36.97      |
| 72.28        | 4337    | 3.43    | 0.0109    | 8.84       |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.1766       | 0.1159  | 0.1039  | 0.0834    |            |

PLOT 7

FIGURE 19

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## FIGURE 20 DESCRIPTION

### SPECTRAL NARROWBAND ENVELOPE ALARM REPORT

The **SPECTRAL NARROWBAND ENVELOPE ALARM REPORT** only prints out spectra for those points in which at least one peak has equaled or exceeded a narrowband envelope alarm.

Narrowband envelope alarms are established by the PMP software based on input of the analyst after at least one set of spectra have been captured for a machine. Currently, some PMP software requires a machine to be in good operating condition when these are set. In these types of software systems, the narrowband alarms are then calculated by setting the envelope a certain percentage above each frequency (for example, setting the envelope 50% to 100% above the amplitude of each peak in the baseline spectrum when the machine is in good operating condition).

Other software offerings allow the user to take in large numbers of similar machines into a "family" and simultaneously create narrowband alarms for the whole machine "family". Of course, some machines will likely be in very good condition, some in very bad shape and the remaining in between these extremes. Therefore, these types of software systems allow the user to go ahead and group large numbers of machines into such a "family" and then employ statistical analysis to have the computer automatically create the envelopes. In these cases this is done by having the software compute the average level of each peak on each spectra for every survey performed, and for every point on each PMP machine. Of course, if the user desires, he can have the software create a different envelope alarm for each position and direction. That is, one envelope setup could be created for position 1A (axial) on each outboard motor bearing in the "family"; then another envelope alarm for position 1H (horizontal) on each machine, etc. Figure 21 through 27 following the **SPECTRAL NARROWBAND ALARM REPORT** are example spectra having at least one envelope in alarm (these spectra are the same Plots 1 thru 7 to which the **RANK-ORDERED RESULTS AND RECOMMENDATIONS REPORT** of Figure 8 referred).

The following columns appear on the report with an explanation of their contents:

| <u>COLUMN</u> | <u>TEXT</u>         | <u>DESCRIPTION</u>                                                                                                      |
|---------------|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1             | (Date)              | Measurement Date                                                                                                        |
| 2             | (Point ID)          | Position Number Where Measurement was Taken and Direction of Probe - Either Axial (A), Horizontal (H), or Vertical (V). |
| 3             | (N'Band ALM-1)      | Narrowband Alarm Name that was used to determine if an Alarm has occurred.                                              |
| 4             | (Over)              | "YES" in this column signifies that at least one Narrowband Envelope Alarm was exceeded.                                |
| 5             | (PK Freq.)          | Frequency with the Highest Amplitude in Spectrum for this Point                                                         |
| 6             | (PK Amp.)           | Amplitude of the Peak Frequency (mils, in/sec, g, gSE, etc.)                                                            |
| 7             | (Units)             | Units of Vibration, whether in Velocity (In/Sec), Displacement (Mils), Acceleration (g), Spike Energy (gSE).            |
| 8             | (OA)                | Amplitude of Current Overall Vibration Measurement *.                                                                   |
| 9             | (OA/ALM)            | Predetermined Overall Alarm Level                                                                                       |
| 10            | (% ALM)             | Percentage of Overall Warning Alarm (100% = At Alarm Level)                                                             |
| 11            | (% CHG)             | Percentage Change of Overall since Previous Survey.                                                                     |
| 12            | (F <sub>MAX</sub> ) | Maximum Frequency Specified when the Data was Collected                                                                 |
| 13            | (Point)             | Point Number assigned by the PMP Software to that Particular Measurement Point                                          |

\* - Overall vibration level here is calculated from the time waveform collected at the time of measurement. In this data collection used, it includes the frequency range of 300 to 3,900,000 CPM ( 5 to 65,000 Hz) independent of the frequency span specified. It is important to note that this is not the value which has been calculated from the spectral data collected which would only include vibration data from 0 CPM to F<sub>MAX</sub> at the time of data collection.

SPECTRAL NARROWBAND ALARM REPORT

| DATE                      | POINT ID   | N'BAND ALM-1 OVER | PK FREQ | PK AMPL | UNITS | OA    | OA/ALM | %ALM | %CHG | FMAX    | POINT |
|---------------------------|------------|-------------------|---------|---------|-------|-------|--------|------|------|---------|-------|
| MACHINE : 102B SUPY AH-44 |            |                   |         |         |       |       |        |      |      |         |       |
| 12/08/92                  | POS 1 - A  | WARNINGLIST YES   | 1080.00 | 1.73312 | IN/S  | 1.740 | 0.4250 | 409  | 32   | 72000.0 | 01400 |
| 12/08/92                  | POS 1 - V  | WARNINGLIST YES   | 1080.00 | 1.30611 | IN/S  | 0.950 | 0.4250 | 224  | 63   | 72000.0 | 01404 |
| 12/08/92                  | POS 2 - A  | WARNINGLIST YES   | 1080.00 | 1.38459 | IN/S  | 1.510 | 0.4250 | 355  | 17   | 72000.0 | 01405 |
| 12/08/92                  | POS 2 - V  | WARNINGLIST YES   | 1800.00 | 0.54504 | IN/S  | 0.710 | 0.4250 | 167  | -18  | 72000.0 | 01409 |
| 12/08/92                  | POS 3 - H  | WARNINGLIST YES   | 1080.00 | 0.67434 | IN/S  | 0.594 | 0.4250 | 140  | 24   | 54000.0 | 01411 |
| 12/08/92                  | POS 3BF    | WARNINGLIST YES   | 1095.00 | 0.55536 | IN/S  | 0.740 | 0.4250 | 174  | 80   | 6000.00 | 01517 |
| MACHINE : CAFETER AHU-308 |            |                   |         |         |       |       |        |      |      |         |       |
| 12/08/92                  | POS 1 - H  | WARNINGLIST YES   | 1800.00 | 0.24694 | IN/S  | 0.315 | 0.4250 | 74   | 21   | 72000.0 | 00515 |
| 12/08/92                  | POS 1 - V  | WARNINGLIST YES   | 1800.00 | 0.71374 | IN/S  | 0.786 | 0.4250 | 185  | 65   | 72000.0 | 00517 |
| 12/08/92                  | POS 2BF    | WARNINGLIST YES   | 315.00  | 0.09665 | IN/S  | 0.171 | 0.4250 | 40   | -10  | 6000.00 | 01524 |
| MACHINE : COND WTR VPMP25 |            |                   |         |         |       |       |        |      |      |         |       |
| 12/08/92                  | POS 1A - A | WARNINGLIST YES   | 25350.0 | 0.14861 | IN/S  | 0.288 | 0.2000 | 144  | 62   | 60000.0 | 00661 |
| 12/08/92                  | POS 3A - A | WARNINGLIST YES   | 49050.0 | 0.34154 | IN/S  | 0.537 | 0.2000 | 269  | 28   | 60000.0 | 00669 |

FIGURE 20

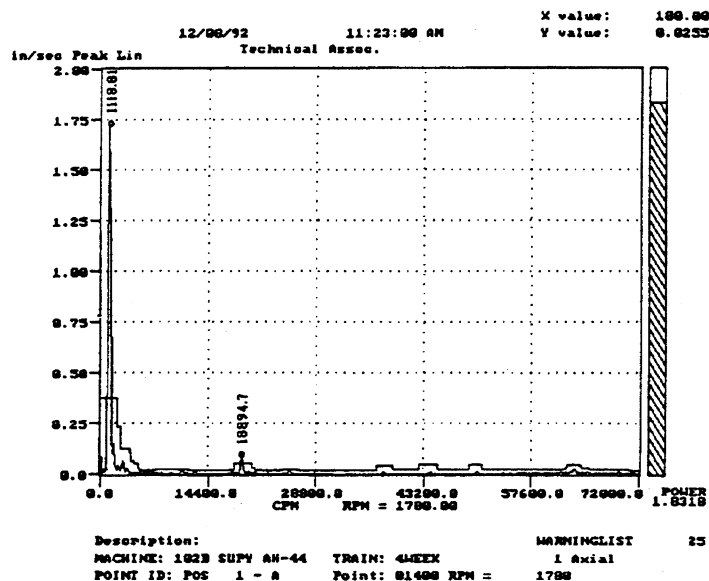
---

## FIGURES 21 THRU 27 DESCRIPTION GRAPHICS REPORT FOR SPECTRA HAVING NARROWBAND ENVELOPE ALARMS

### A. Vibration Spectra

This section includes plots of Vibration Amplitude versus Frequency (with Narrowband Spectral Envelope Bands displayed on them). Spectra are included for those machines which have either experienced significant vibration increases, are now in **ALARM** as determined by the **NARROWBAND ALARM REPORT**, or which have individual frequencies indicating duress of a particular machine component(s) whether or not the amplitude of these particular frequencies have increased overall levels into **ALARM** severity. Where applicable, additional notation is provided on the plot relating information on problem components and/or conditions which have been detected. Machine name, point location, vibration parameter, amplitude scale and frequency range are provided on each plot, along with the narrowband envelope alarms.

Figures 21 thru 27 (Plots 1 thru 7) are the plots of the spectra from the **RANK-ORDERED SUMMARY OF RESULTS AND RECOMMENDATIONS FOR PROBLEM MACHINES** (See Figure 8) which were machines having been diagnosed as having problems. Note that the "Plot Nos." were referred to in the last column of this report (See Figure 8) so that the reader can compare the diagnosed problem to the spectra from which the diagnosis was made.



12/08/92  
11:23:00 AM

### Threshold Report

PLANT:  
Description:

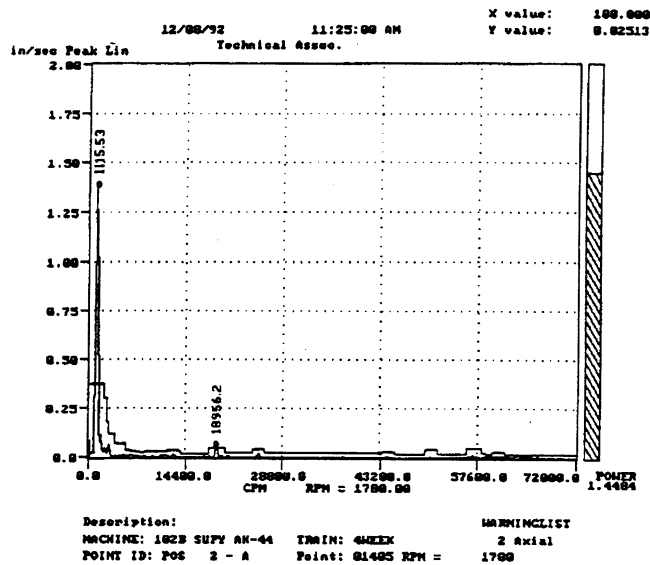
MACHINE: 102B SUPY AH-44  
POINT ID: POS 1 - A  
Units: in/sec Peak Lin  
Threshold: 0.0100

TRAIN: 4WEEK  
Point: 01400  
RPM: 1780.00

| Frequency Hz                | CPM     | Order  | Spect Amp | % Exceeded |
|-----------------------------|---------|--------|-----------|------------|
| 3.00                        | 180     | 0.10   | 0.0255    | 154.53     |
| 18.65                       | 1.12E+3 | 0.63   | 1.7331    | 17231.20   |
| 48.97                       | 2938    | 1.65   | 0.0629    | 528.84     |
| 58.62                       | 3517    | 1.98   | 0.0279    | 178.62     |
| 180.98                      | 1.09E+4 | 6.10   | 0.0200    | 99.98      |
| 210.00                      | 12600   | 7.08   | 0.0157    | 57.12      |
| 314.91                      | 1.89E+4 | 10.62  | 0.0959    | 859.07     |
| 330.57                      | 19834   | 11.14  | 0.0103    | 3.02       |
| 343.96                      | 2.06E+4 | 11.59  | 0.0103    | 3.02       |
| 419.91                      | 2.52E+4 | 14.15  | 0.0247    | 146.97     |
| 944.56                      | 56673   | 31.84  | 0.0103    | 3.02       |
| 1038.36                     | 6.23E+4 | 35.00  | 0.0120    | 19.78      |
| 1056.77                     | 6.34E+4 | 35.62  | 0.0305    | 2.05E+2    |
| 1074.95                     | 64497   | 36.23  | 0.0103    | 3.02       |
| 1155.23                     | 6.93E+4 | 38.94  | 0.0103    | 3.02       |
| Power Sync Subsync Non-Sync |         |        |           |            |
| 1.8318                      | 0.1592  | 1.8194 | 0.1404    |            |

PLOT 1

FIGURE 21



12/08/92  
11:25:00 AM

### Threshold Report

PLANT:  
Description:

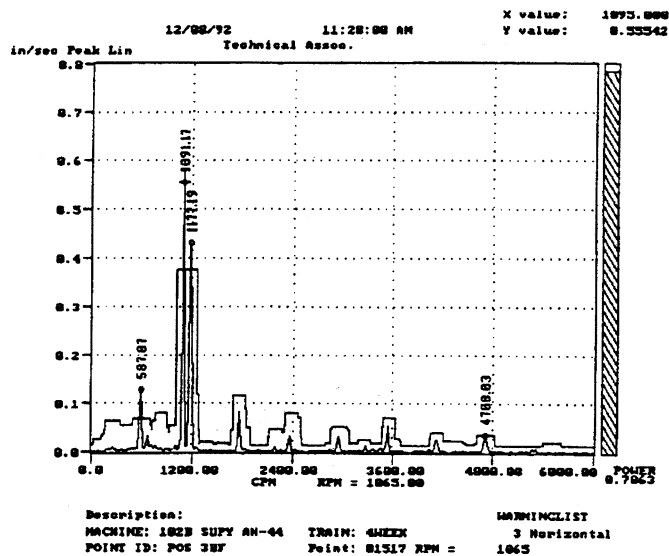
MACHINE: 102B SUPY AH-44  
POINT ID: POS 2 - A  
Units: in/sec Peak Lin  
Threshold: 0.0100

TRAIN: 4WEEK  
Point: 01405  
RPM: 1780.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 3.00         | 180     | 0.10    | 0.0251    | 151.29     |
| 18.59        | 1.12E+3 | 0.63    | 1.3846    | 13745.90   |
| 39.32        | 2359    | 1.33    | 0.0485    | 384.85     |
| 49.24        | 2954    | 1.66    | 0.0673    | 573.48     |
| 105.91       | 6.35E+3 | 3.57    | 0.0130    | 30.24      |
| 181.16       | 10869   | 6.11    | 0.0122    | 21.95      |
| 190.60       | 11436   | 6.42    | 0.0107    | 6.93       |
| 210.70       | 1.26E+4 | 7.10    | 0.0206    | 106.32     |
| 315.94       | 18956   | 10.65   | 0.0652    | 551.71     |
| 421.25       | 25275   | 14.20   | 0.0228    | 127.70     |
| 947.42       | 56845   | 31.94   | 0.0122    | 21.95      |
| 1052.74      | 6.32E+4 | 35.49   | 0.0153    | 53.50      |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 1.4484       | 0.1202  | 1.4380  | 0.1241    |            |

PLOT 2

FIGURE 22



12/08/92  
11:28:00 AM

### Threshold Report

PLANT:  
Description:

MACHINE: 102B SUPY AH-44

POINT ID: POS 3BF

Units: in/sec Peak Lin

Threshold: 0.0100

TRAIN: 4WEEK

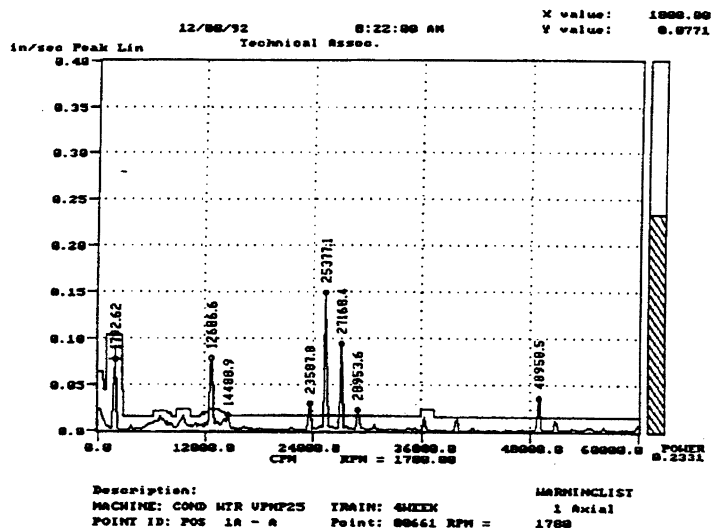
Point: 01517

RPM: 1065.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 4.20         | 2.52E+2 | 0.24    | 0.0112    | 12.00      |
| 9.80         | 5.88E+2 | 0.55    | 0.1275    | 1174.51    |
| 11.17        | 670     | 0.63    | 0.0322    | 221.96     |
| 11.92        | 7.15E+2 | 0.67    | 0.0149    | 49.38      |
| 12.77        | 766     | 0.72    | 0.0136    | 35.71      |
| 16.72        | 1.00E+3 | 0.94    | 0.0149    | 49.38      |
| 18.19        | 1091    | 1.02    | 0.5554    | 5453.61    |
| 19.62        | 1177    | 1.11    | 0.4299    | 4199.37    |
| 20.85        | 1251    | 1.17    | 0.0102    | 1.75       |
| 29.27        | 1756    | 1.65    | 0.0841    | 740.80     |
| 36.39        | 2183    | 2.05    | 0.0108    | 8.47       |
| 39.31        | 2.36E+3 | 2.21    | 0.0332    | 232.43     |
| 49.15        | 2949    | 2.77    | 0.0332    | 232.43     |
| 54.59        | 3.28E+3 | 3.08    | 0.0140    | 40.12      |
| 59.01        | 3.54E+3 | 3.32    | 0.0555    | 454.67     |
| 68.65        | 4.12E+3 | 3.87    | 0.0266    | 165.72     |
| 78.47        | 4708    | 4.42    | 0.0366    | 265.92     |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.7863       | 0.3101  | 0.1394  | 0.7090    |            |

PLOT 3

FIGURE 23



12/08/92  
8:22:00 AM

### Threshold Report

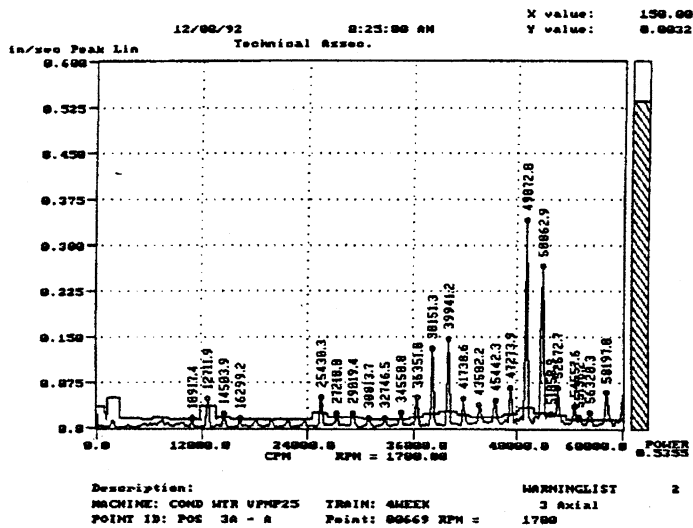
PLANT:  
Description:  
MACHINE: COND WTR VPMP25  
POINT ID: POS 1A - A  
Units: in/sec Peak Lin  
Threshold: 0.0100

TRAIN: 4WEEK  
Point: 00661  
RPM: 1780.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 2.50         | 150     | 0.08    | 0.0241    | 141.48     |
| 29.88        | 1.79E+3 | 1.01    | 0.0771    | 671.03     |
| 103.65       | 6219    | 3.49    | 0.0105    | 5.00E+0    |
| 113.56       | 6.81E+3 | 3.83    | 0.0165    | 65.38      |
| 127.50       | 7650    | 4.30    | 0.0102    | 2.38       |
| 158.53       | 9.51E+3 | 5.34    | 0.0170    | 69.60      |
| 211.44       | 1.27E+4 | 7.13    | 0.0791    | 690.74     |
| 241.48       | 1.45E+4 | 8.14    | 0.0170    | 69.60      |
| 393.13       | 2.36E+4 | 13.25   | 0.0296    | 195.51     |
| 422.95       | 25377   | 14.26   | 0.1486    | 1386.10    |
| 452.81       | 27168   | 15.26   | 0.0944    | 843.53     |
| 482.56       | 2.90E+4 | 16.27   | 0.0224    | 123.87     |
| 604.34       | 36260   | 20.37   | 0.0153    | 53.32      |
| 664.11       | 3.98E+4 | 22.39   | 0.0128    | 28.49      |
| 815.98       | 4.90E+4 | 27.50   | 0.0362    | 261.62     |
| 845.93       | 5.08E+4 | 28.51   | 0.0119    | 19.12      |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.2332       | 0.1025  | 0.0321  | 0.2069    |            |

PLOT 4

FIGURE 24



12/08/92  
8:25:00 AM

### Threshold Report

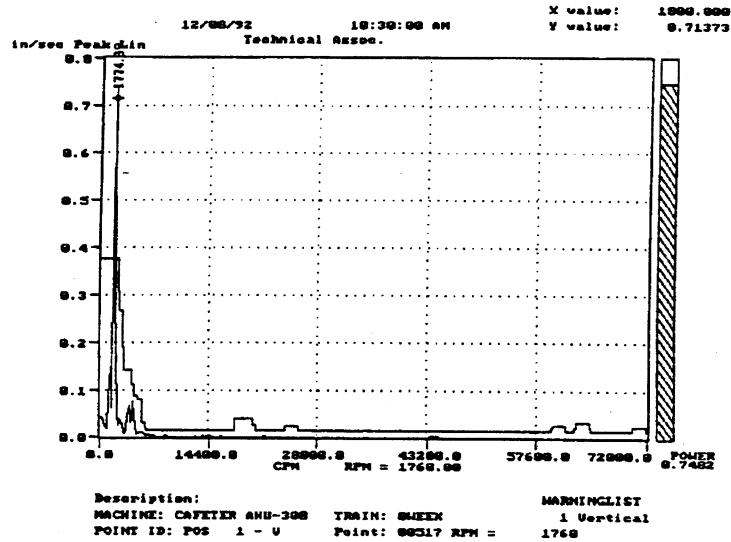
PLANT:  
Description:  
MACHINE: COND WTR VPMP25  
POINT ID: POS 3A - A  
Units: in/sec Peak Lin  
Threshold: 0.0200

TRAIN: 4WEEK  
Point: 00669  
RPM: 1780.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 211.87       | 1.27E+4 | 7.14    | 0.0489    | 144.55     |
| 241.73       | 1.45E+4 | 8.15    | 0.0234    | 16.79      |
| 423.84       | 25430   | 14.29   | 0.0503    | 151.34     |
| 453.65       | 2.72E+4 | 15.29   | 0.0240    | 20.03      |
| 483.66       | 29019   | 16.30   | 0.0234    | 16.79      |
| 575.98       | 3.46E+4 | 19.42   | 0.0254    | 26.78      |
| 605.85       | 36351   | 20.42   | 0.0503    | 151.34     |
| 635.86       | 38151   | 21.43   | 0.1310    | 555.14     |
| 665.69       | 39941   | 22.44   | 0.1462    | 630.95     |
| 695.64       | 4.17E+4 | 23.45   | 0.0476    | 137.95     |
| 726.37       | 43582   | 24.48   | 0.0372    | 85.99      |
| 757.37       | 45442   | 25.53   | 0.0438    | 119.19     |
| 787.90       | 4.73E+4 | 26.56   | 0.0661    | 230.48     |
| 807.36       | 48441   | 27.21   | 0.0204    | 1.85       |
| 817.88       | 4.91E+4 | 27.57   | 0.3415    | 1607.70    |
| 835.82       | 50149   | 28.17   | 0.0209    | 4.68       |
| 847.72       | 5.09E+4 | 28.57   | 0.2670    | 1234.82    |
| 864.32       | 5.19E+4 | 29.13   | 0.0240    | 20.03      |
| 877.88       | 5.27E+4 | 29.59   | 0.0661    | 230.48     |
| 909.29       | 5.46E+4 | 30.65   | 0.0362    | 80.97      |
| 938.67       | 56320   | 31.64   | 0.0254    | 26.78      |
| 969.96       | 5.82E+4 | 32.70   | 0.0592    | 196.20     |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.5373       | 0.0711  | 0.0056  | 0.5326    |            |

PLOT 5

FIGURE 25



12/08/92  
10:30:00 AM

### Threshold Report

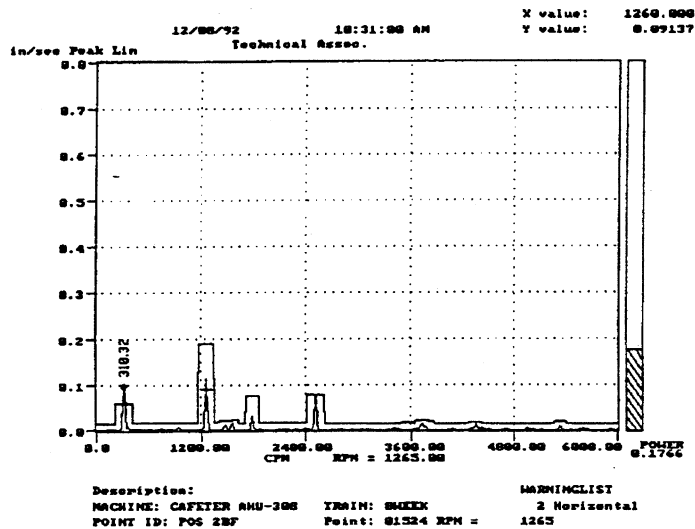
PLANT:  
Description:

MACHINE: CAFETER AHU-308  
POINT ID: POS 1 - V  
Units: in/sec Peak Lin  
Threshold: 0.0100

TRAIN: 8WEEK  
Point: 00517  
RPM: 1760.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 3.00         | 180     | 0.10    | 0.0419    | 318.51     |
| 29.57        | 1774    | 1.01    | 0.7137    | 7037.40    |
| 62.27        | 3736    | 2.12    | 0.0664    | 564.10     |
| 72.21        | 4.33E+3 | 2.46    | 0.0758    | 657.76     |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.7482       | 0.7237  | 0.1579  | 0.1058    |            |

PLOT 6  
FIGURE 26



12/08/92  
10:31:00 AM

### Threshold Report

PLANT:

Description:

MACHINE: CAFETER AHU-308

POINT ID: POS 2BF

Units: in/sec Peak Lin

Threshold: 0.0100

TRAIN: 8WEEK

Point: 01524

RPM: 1265.00

| Frequency Hz | CPM     | Order   | Spect Amp | % Exceeded |
|--------------|---------|---------|-----------|------------|
| 5.17         | 310     | 0.25    | 0.0966    | 866.48     |
| 20.91        | 1.25E+3 | 0.99    | 0.0913    | 812.51     |
| 24.43        | 1.47E+3 | 1.16    | 0.0112    | 12.01      |
| 25.81        | 1.55E+3 | 1.22    | 0.0167    | 67.49      |
| 29.64        | 1.78E+3 | 1.41    | 0.0315    | 215.15     |
| 41.87        | 2.51E+3 | 1.99    | 0.0790    | 690.39     |
| 61.94        | 3716    | 2.94    | 0.0137    | 36.97      |
| 72.28        | 4337    | 3.43    | 0.0109    | 8.84       |
| Power        | Sync    | Subsync | Non-Sync  |            |
| 0.1766       | 0.1159  | 0.1039  | 0.0834    |            |

PLOT 7

FIGURE 27

---

## FIGURE 28 DESCRIPTION CURRENT INSPECTION CODE REPORT

The **INSPECTION CODE REPORT** is a listing of inspection codes stored during the survey with the corresponding machine and other applicable information. These inspection codes are entered and stored in the field when abnormalities in spectra or operating conditions are noted. Although these codes will appear in both the **ACTION** and the **MACHINERY MEASUREMENT** reports, the **CURRENT INSPECTION CODE REPORT** is included for a "quick reference" to alert maintenance personnel of conditions considered abnormal as observed during the survey.

Page No. 1  
01/05/93

### I N S P E C T I O N   C O D E   R E P O R T

| TE       | POINT ID          | INSPECTION REMARKS         |
|----------|-------------------|----------------------------|
| MACHINE  | : 102B SUPY AH-44 |                            |
| 01/05/93 | POS 1 - H         | Fluctuating Vibration      |
| 01/05/93 | POS 2-H HI BPF    | Belts Loose or Missing     |
| MACHINE  | : CAFETER AHU-308 |                            |
| 01/05/93 | POS 4 - H         | Safety Condition Indicated |
| 01/05/93 | POS 1 - H         | Loose Parts                |
| 01/05/93 | POS 3 - H         | Abnormal Noise             |
| MACHINE  | : COND WTR VPMP25 |                            |
| 01/05/93 | POS 1A - A        | Leaks                      |
| 01/05/93 | POS 3W - O        | Fluctuating Vibration      |
| 01/05/93 | POS 1N - O        | Hot Bearing                |

FIGURE 28

---

## FIGURES 29 THRU 31 DESCRIPTION TRENDS AND WATERFALL PLOTS

Figure 29 shows 3 different plots. The top plot is a trend of Overall Velocity starting on 4/18/91 and progressing to the current measurement at 12/8/92 with data taken on the inboard pump bearing (vertical pump). Note that it shows a marked increase in overall velocity amplitude, but that no diagnostics can be performed from this plot alone. It merely tells the reader that something is wrong since the alarm levels (shown on the plot) have been exceeded (both ALARM 1 and ALARM 2)

The middle plot of Figure 29 show a Velocity Spectrum with narrowband envelope alarms for the latest (12/8/92) measurement in orders (orders are multiples of running speed; i.e., if a machine is running at 3580 RPM, the 1st order would be at 3580 CPM, the 2nd order would be at 7160 CPM, and so on . . . ). Note that the 7.14 order of running speed peak has exceeded the narrowband alarm. This 7.14X RPM frequency correlates with the Bearing Inner Race Defect Frequency Multiplier (BPFI). Also, note that many harmonics of the BPFI order have appeared (along with many IX RPM Sideband Frequencies), and are labeled in this plot. This is a spectrum indicating significant bearing wear which must be replaced immediately.

The bottom plot of Figure 29 is a trend of Overall Spike Energy (gSE) from 4/18/91 to 12/8/92 from the same pump inboard bearing. Notice how the spike energy level has been increasing over the past several measurements. Also, note that the pattern was somewhat erratic which is common with spike energy measurements. This increase in spike energy levels can be attributed to many factors which include bearing wear, lubrication problems, cavitation, rubs, etc. However, it is necessary to view the spectrum (middle of Figure 29) to determine exactly what the problem is.

The top of Figure 30 is a waterfall plot (also known as a "Cascade Plot") on the same inboard pump bearing (again in the axial direction) that was found to have a problem in Figure 29. A waterfall plot is a date-ordered representation of the frequency spectral history for this point. Note the dates on the right-hand side of the figure. Also, note the appearance, and then the growth of inner race frequencies for this inboard pump bearing. Analysis of the bottom spectrum in Figure 30 (from the 12/8/92 data) will clearly show these bearing defect frequencies.

Figure 31 is another waterfall plot, but taken in the axial direction on the top (outboard) motor bearing housing of this vertical pump installation (a distance of approximately 6 to 7 feet from the inboard pump bearing). Note even here on the other side of the coupling on the upper motor bearing that the pump bearing problem was detected during the same 6/9/92 survey. Here again, more pump bearing inner race frequency harmonics appeared and grew until the latest survey of 12/8/92. A strong recommendation was made to replace this inboard pump bearing ASAP.

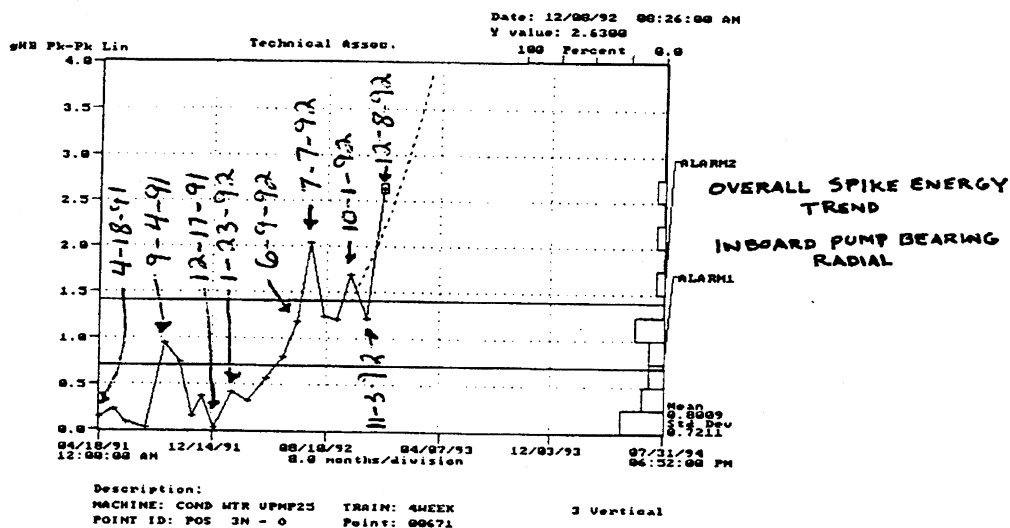
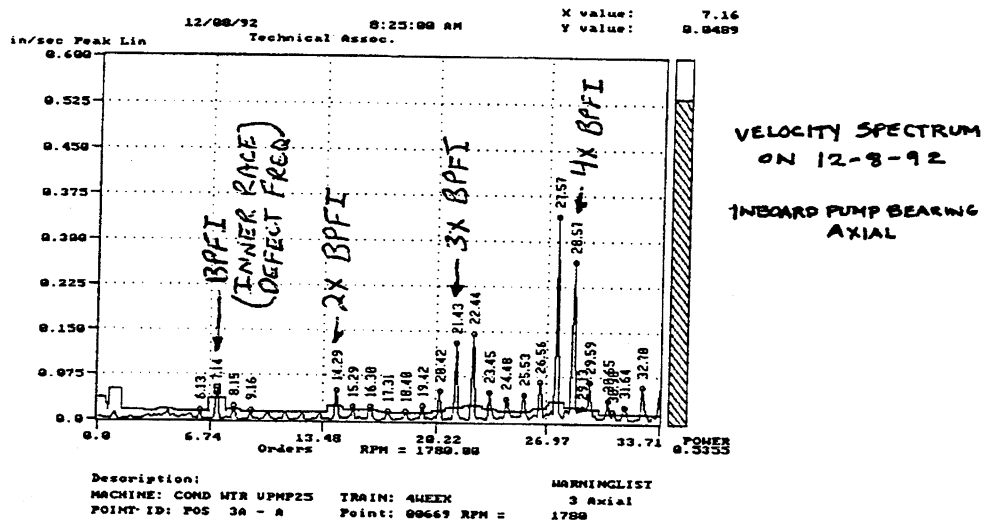
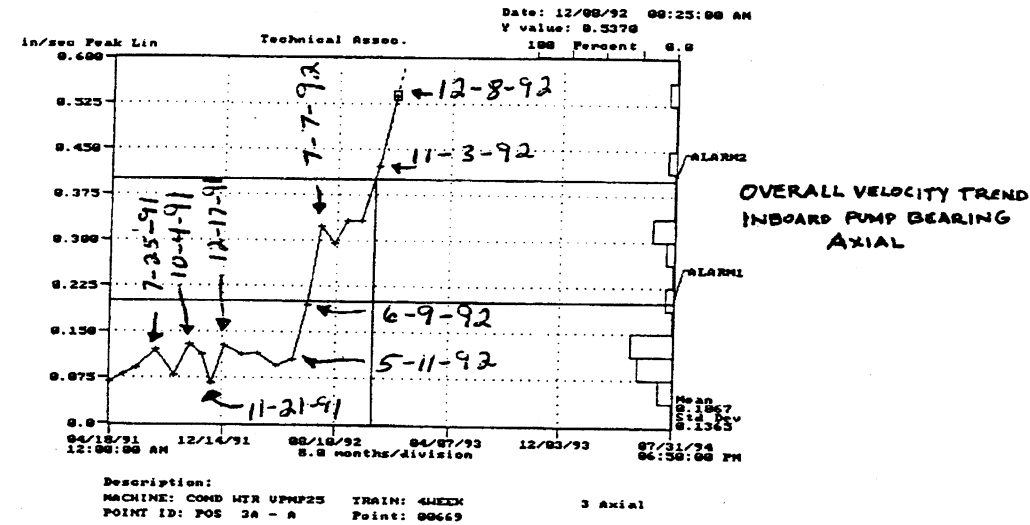
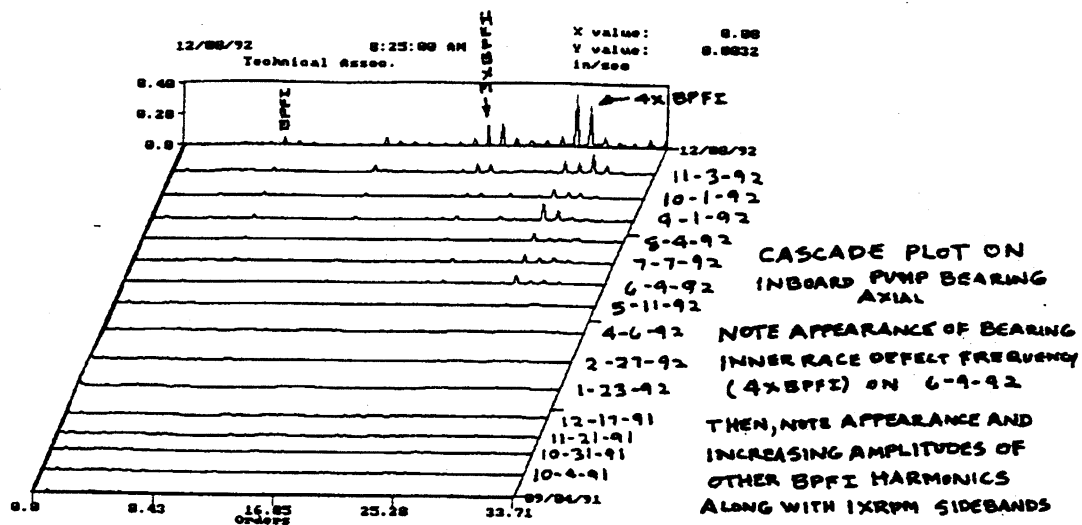


FIGURE 29



Description:  
MACHINE: COND MTR UPMF25 TRAIN: 4MEEK RPM = 1780.00  
POINT ID: POS 3a - a Point: 80669 RPM = 1780 3 Axial

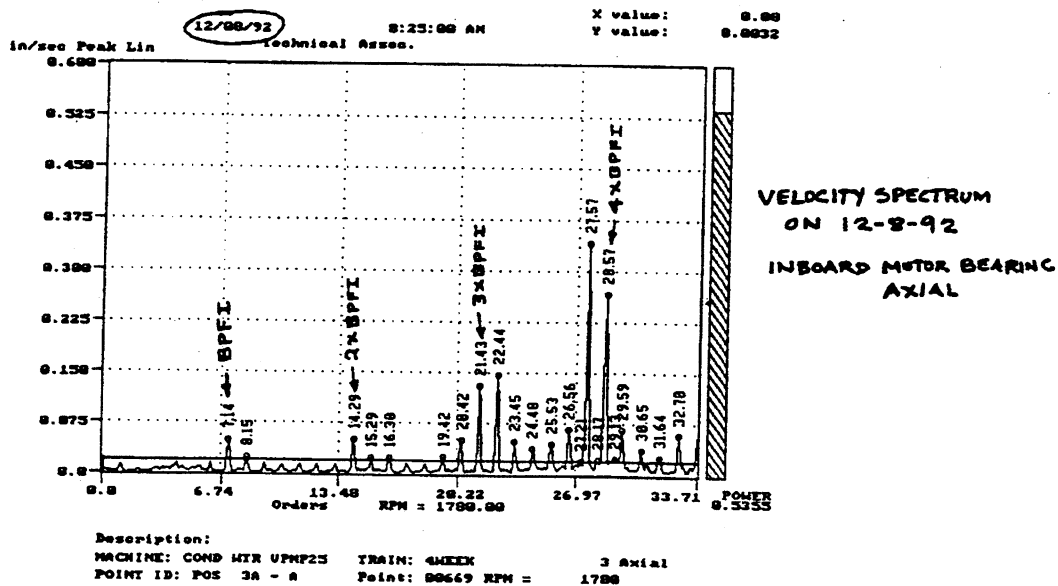
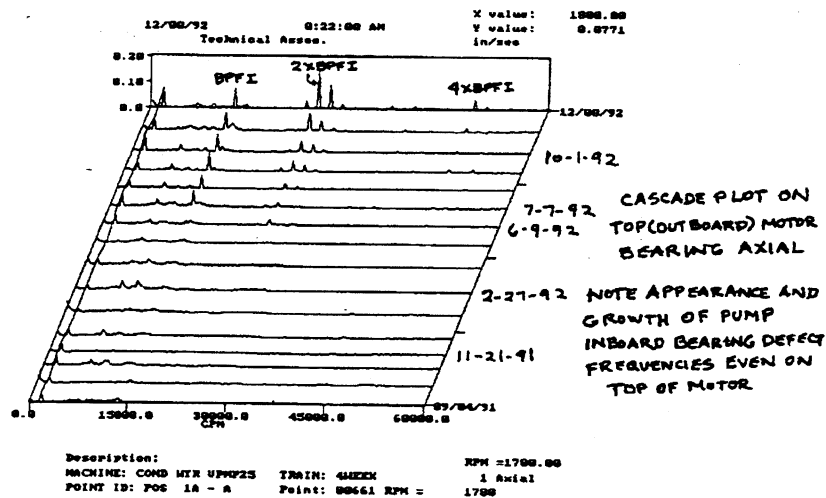


FIGURE 30



12/08/92  
 8:22:00 AM

### Threshold Report

PLANT:  
 Description:  
 MACHINE: COND WTR VPMP25 12-8-92 TRAIN: 4WEEK  
 POINT ID: POS 1A - A DATA Point: 00661  
 Units: in/sec Peak Lin TOP MOTOR BEARING RPM: 1780.00  
 Threshold: 0.0100

| Frequency Hz | CPM     | Order         | Spect Amp | % Exceeded |
|--------------|---------|---------------|-----------|------------|
| 2.50         | 150     | 0.08          | 0.0241    | 141.48     |
| 29.88        | 1.79E+3 | 1.01          | 0.0771    | 671.03     |
| 103.65       | 6219    | 3.49          | 0.0105    | 5.00E+0    |
| 113.56       | 6.81E+3 | 3.83          | 0.0165    | 65.38      |
| 127.50       | 7650    | 4.30          | 0.0102    | 2.38       |
| 158.53       | 9.51E+3 | 5.34          | 0.0170    | 69.60      |
| 211.44       | 1.27E+4 | 7.13 ← BPFI   | 0.0791    | 690.74     |
| 241.48       | 1.45E+4 | 8.14          | 0.0170    | 69.60      |
| 393.13       | 2.36E+4 | 13.25         | 0.0296    | 195.51     |
| 422.95       | 25377   | 14.26 ← 2BPFI | 0.1486    | 1386.10    |
| 452.81       | 27168   | 15.26         | 0.0944    | 843.53     |
| 482.56       | 2.90E+4 | 16.27         | 0.0224    | 123.87     |
| 604.34       | 36260   | 20.37         | 0.0153    | 53.32      |
| 664.11       | 3.98E+4 | 22.39         | 0.0128    | 28.49      |
| 815.98       | 4.90E+4 | 27.50         | 0.0362    | 261.62     |
| 845.93       | 5.08E+4 | 28.51 ← 4BPFI | 0.0119    | 19.12      |
| Power        | Sync    | Subsync       | Non-Sync  |            |
| 0.2332       | 0.1025  | 0.0321        | 0.2069    |            |

FIGURE 31

---

## EFFECT OF VIBRATION ACCEPTANCE TESTING

### EFFECTS ON NEW AND REBUILT MOTOR COSTS AND MAINTENANCE CREW SIZE

(Excerpt from Practical Solution to Machinery and Maintenance Vibration Problems: Ralph T. Buscarello; Update International, Inc.,  
6320 West Lakeridge Road, Denver, CO 80227)

*During harsh economic times, companies are often forced to maintain the same number of machines with less people. During times of expansion, they often have to add more people. But look at how vibration control can affect maintenance crew size, to save in both harsh as well as good times. A chemical plant with 1700 rotating machine assemblies driven by 1200 electric motors reported the following results:*

*From mid 1977 through mid 1980, they reduced their motor rewinding costs from \$172,000/year (1977-1978) to \$40,000/year (1979-1980), for a \$132,000 savings on that item alone. Each new or rebuilt motor delivered underwent a vibration acceptance test, including rolling element bearing analysis. A surprising 14% of new motors were rejected mostly for such bearing related problems as improper assembly or clearances, too tight a press fit, etc. They reported: "If we had put these motors in the field, they would have failed within 3 to 6 months." By finding a failing bearing or a coupling beginning to bind, overheating and machine failure was prevented, thus eliminating the need for rewinding. Vibration crews said: "The key result is increased lead time so we can properly schedule and have the right parts and people available on a no-panic basis." Before the program, operators wrote orders only when machines failed. Under the vibration control program, they now call for machine analysis whenever a change occurs, usually long before failure. If correction is not immediately possible, life prediction enables proper scheduling for shutdown and repair.*

*During that same period, the plant was expanded and the company increased its number of rotating machines by 41.8%. Ordinarily, such expansion would have required 20 more maintenance personnel. However, because of diligent vibration control, the number of millwright/mechanics was actually decreased from 63 in 1977 to 43 in 1980. Consider this in the light of a down economy which may require maintaining the same amount of machinery with a smaller crew.*

*This could give you the idea that we're in favor of putting people out of work. It is very far from the truth. Our conviction is that where plants are less efficient, less people have jobs. More productive plants almost always result in the kind of growth and healthy working conditions that produce more and more jobs, as well as profits. Throughout the world, the less efficient industries are constantly being replaced by the more efficient ones. Our company favors the worker by helping their companies eliminate waste - the kind of waste that drains away jobs as well as profits.*

### SPECIFIC EFFECTS ON MAINTENANCE COSTS IN DOLLARS PER HORSEPOWER

*A medium sized oil refinery in Mississippi published a detailed financial report in the Oil & Gas Journal of October 18, 1976, documenting their maintenance cost reductions through their 5 year old vibration control program. This company had a unique way of measuring returns in "maintenance costs per horsepower." For 100 major machines, totaling 165,000 hp, annual costs were reduced from \$7 per hp to \$5 per hp. About 3900 "minor" machines (1,000 hp on down) such as ordinary pumps, motors, fans, etc., showed reduction of approximately 30% from \$6.35 to \$4.47 per hp per year. These "minor" machines totaled 340,000 hp. Simple arithmetic shows maintenance savings alone of almost \$900,000 per year. Yet these maintenance savings were undoubtedly small compared to the extra production income.*

---

## **PRACTICAL IN-DEPTH VIBRATION TRAINING**

*The best equipment in the world is almost useless if those who use it have no practical, in-depth training. Vibration training has to extend considerably beyond the instruments themselves and get into the "nuts and bolts" of just what is required to prevent, predict and correct almost any vibration problem that may arise. This cannot be done with a day of fast training by the local salesman. A practical "nuts and bolts" approach allows any person with a reasonable understanding of his plant's machinery and a fairly good reasoning ability, to readily learn all that he needs to know to carry out all phases of vibration analysis, prevention, and correction.*

### **INTERPLAY BETWEEN SUPERVISORS, ENGINEERS, FOREMEN, CRAFTSMEN, AND TECHNICIANS**

*Another resistance to getting the proper financial returns is when all vibration control is only at a higher technical or engineering level, and not on the practical level of those closest to the machinery itself. Even though not everyone on the machinery team is full time in vibration control, vibration knowledge is important for:*

|                              |                                                         |
|------------------------------|---------------------------------------------------------|
| <i>Maintenance Engineers</i> | <i>Area Supervisors</i>                                 |
| <i>Inspectors</i>            | <i>Rotating Equipment Specialists</i>                   |
| <i>Reliability Engineers</i> | <i>Shop Supervisors</i>                                 |
| <i>Advanced Mechanics</i>    | <i>Maintenance and Engineering Supervisors/Managers</i> |

*Too often Maintenance and Engineering Managers remain aloof from trying to understand vibration as they feel it is too complicated or technical. They get confused by the technical jargon used. Vibration knowledge can be brought down to the ordinary and practical level, and no supervisor who is responsible for large amounts of very expensive rotating machinery could really ignore or "delegate" the subject without having it show up in his work. A Maintenance Manager who attended one of Update's courses put it this way, "At first, I was sending only technical people who report to me to be trained, and found that at important meetings, they knew more about it than I did. Since I was responsible for machinery decisions that involved many thousands of dollars, sometimes for only one machine, I decided to get the training myself. Now I have a more complete perspective."*

*Then there are those plants or mills that have engineers or technicians doing a low level of vibration work that would soon bore a self-respecting engineer or technically trained person. For example, in the beginning, an engineer may enjoy field balancing. It is like a new toy to him. But after a few routine balancing jobs, he would subconsciously avoid working around the machinery, having "too many other things to do." This leads to actually working on only those situations that are extreme, ignoring those machines that would still run, even if "a little rough." The price is finally paid when the machines wear out bearings or seals that much sooner. The same applies for the routine analysis and procedures that could easily be performed by those that do not have as much education. Upon investigation, a fair percentage of vibration control can be preformed by people without formal higher education. Such people find this work a challenge, bringing their regular work out of the ordinary, less boring, and more importantly, get it done.*